

Distinguishable and Exchangeable Dyads: Bayesian Multilevel Modelling

Cross-Sectional and Intensive Longitudinal APIM and DIM

Pascal M. Küng

2026-06-16

This tutorial presents applied Bayesian multilevel modelling for dyadic data, covering both distinguishable and exchangeable dyads. We develop cross-sectional and intensive longitudinal variants of the Actor–Partner Interdependence Model (APIM) and the Dyad–Individual Model (DIM), demonstrate their algebraic equivalence, and discuss implications for interpretation. Using simulated data, we provide fully reproducible code to rebuild the models and assess parameter recovery. The tutorial offers comprehensive `{brms}` implementations, including random-effects structures, heterogeneous residual variances, compound-symmetry residuals, and optional AR(1) processes. We illustrate diagnostics and model comparison (e.g., posterior predictive checks and LOO), and show how to extract dyad-level variance–covariance components for exchangeable-dyad models. All code is open and includes data simulation routines that can be run to test parameter retrieval. The material serves as a hands-on guide for analyzing dyadic processes in both cross-sectional and longitudinal settings.

Interactive HTML Version (recommended):

This tutorial is best viewed as interactive slides here:

<https://pascal-kueng.github.io/05DyadicDataAnalysis/DyadicDataAnalysis.html>

The PDF version below is a static archival version for indexing, citation, and long-term reference.

Version DOI: 10.5281/zenodo.20709781

All-versions DOI: 10.5281/zenodo.17400655

Suggested citation: Küng, P. M. (2026). *Distinguishable and Exchangeable Dyads: Bayesian Multilevel Modelling* (Version v2.0.5). University of Zurich. DOI: 10.5281/zenodo.20709781

0.1 Overview

- Distinguishable Dyads
 - Cross-Sectional APIM
 - Longitudinal APIM
- Exchangeable Dyads
 - Cross-Sectional APIM
 - Cross-Sectional DIM
 - Equivalence between APIM and DIM

- Longitudinal DIM
- Longitudinal APIM

0.2 Loading Libraries and setting CmdStan Backend

Some functions used in this presentation are sourced from the files available in the folder 00_R_Functions.

```
library(tidyverse)
library(brms)
library(glmTMB)
library(bmlm)
library(easystats)
library(DiagrammeR)
library(DHARMa)
library(kableExtra)
library(MASS)
library(purrr)
source(file.path('00_R_Functions', 'ReportModels.R'))
source(file.path('00_R_Functions', 'PrettyTables.R'))
source(file.path('00_R_Functions', 'PrepareData.R'))

# PDF tables need to be substantially smaller than slide tables.
print_df_slide <- print_df
print_df <- function(df,
                      caption = NULL,
                      digits = NULL,
                      width = NULL,
                      rows_to_pack = NULL,
                      scroll_height = "auto",
                      scroll_width = "100%",
                      font_size = 7) {
  if (!knitr::is_latex_output()) {
    return(print_df_slide(
      df,
      caption = caption,
      digits = digits,
      width = width,
      rows_to_pack = rows_to_pack,
      scroll_height = scroll_height,
      scroll_width = scroll_width,
      font_size = font_size
    ))
  }
  df <- as.data.frame(df)
  if (!identical(rownames(df), as.character(seq_len(nrow(df))))) {
    df <- tibble::rownames_to_column(df, var = " ")
  }
  validate_packing_input(rows_to_pack, nrow(df))

  clean_pdf_label <- function(x) {
    x <- gsub("provided_support", "support", x, fixed = TRUE)
    x <- gsub("communication", "comm", x, fixed = TRUE)
    x <- gsub("diaryday", "day", x, fixed = TRUE)
    x <- gsub("_gmc", "", x, fixed = TRUE)
    x <- gsub("_cbp", " BP", x, fixed = TRUE)
    x <- gsub("_cwp", " WP", x, fixed = TRUE)
    x <- gsub("_cw", " W", x, fixed = TRUE)
    x <- gsub("_cb", " B", x, fixed = TRUE)
    x <- gsub("-", " ", x, fixed = TRUE)
    x <- gsub(":", ":", x, fixed = TRUE)
    x <- gsub(",", " ", x, fixed = TRUE)
    x
  }

  names(df) <- clean_pdf_label(names(df))

  if (!is.null(digits)) {
    numeric_cols <- vapply(df, is.numeric, logical(1))
  }
}
```

```

df[numeric_cols] <- lapply(df[numeric_cols], round, digits = digits)
}

character_cols <- vapply(df, is.character, logical(1))
df[character_cols] <- lapply(df[character_cols], clean_pdf_label)

use_longtable <- nrow(df) > 28
wide_table <- ncol(df) >= 5 ||
  any(vapply(df, function(x) {
    if (!is.character(x)) return(FALSE)
    max(nchar(x), na.rm = TRUE) > 22
  }, logical(1)))

alignments <- ifelse(vapply(df, is.numeric, logical(1)), "r", "l")
tbl <- knitr::kable(
  df,
  caption = caption,
  escape = FALSE,
  align = alignments,
  row.names = FALSE,
  booktabs = TRUE,
  longtable = use_longtable
)

if (use_longtable) {
  tbl <- tbl |>
  kableExtra::kable_styling(
    font_size = min(font_size, 7),
    latex_options = "repeat_header",
    position = "left"
  )
} else {
  latex_options <- character()
  if (wide_table && is.null(rows_to_pack)) {
    latex_options <- "scale_down"
  }
  if (length(latex_options) > 0) {
    tbl <- suppressWarnings(
      tbl |>
      kableExtra::kable_styling(
        font_size = font_size,
        latex_options = latex_options,
        position = "left"
      )
    )
  } else {
    tbl <- tbl |>
    kableExtra::kable_styling(
      font_size = font_size,
      position = "left"
    )
  }
}

if (use_longtable && ncol(df) >= 4) {
  tbl <- tbl |>
  kableExtra::column_spec(1, width = "13em")
}

packing(tbl, rows_to_pack)
}

print_wide_matrix <- function(x, digits = 3, font_size = 6) {
  df <- as.data.frame(round(x, digits))
  if (!identical(row.names(df), as.character(seq_len(nrow(df))))) {
    df <- tibble::rownames_to_column(df, var = " ")
  }

  clean_pdf_label <- function(x) {
    x <- gsub("provided_support", "support", x, fixed = TRUE)
    x <- gsub("communication", "comm", x, fixed = TRUE)
    x <- gsub("diaryday", "day", x, fixed = TRUE)
    x <- gsub("PartnerA", "PA", x, fixed = TRUE)
    x <- gsub("PartnerB", "PB", x, fixed = TRUE)
  }

```

```

x <- gsub("Intercept", "Int", x, fixed = TRUE)
x <- gsub("support", "sup", x, fixed = TRUE)
x <- gsub("actor", "act", x, fixed = TRUE)
x <- gsub("partner", "prt", x, fixed = TRUE)
x <- gsub("_", " ", x, fixed = TRUE)
x <- gsub(":", " ", x, fixed = TRUE)
x <- gsub(",", " ", x, fixed = TRUE)
x
}

names(df) <- clean_pdf_label(names(df))
df <- df |> dplyr::mutate(dplyr::across(where(is.character), clean_pdf_label))

suppressWarnings(
  knitr::kable(
    df,
    escape = FALSE,
    row.names = FALSE,
    booktabs = TRUE,
    align = ifelse(vapply(df, is.numeric, logical(1)), "r", "l")
  ) |>
  kableExtra::kable_styling(
    font_size = font_size,
    latex_options = "scale_down",
    position = "left"
  )
)
}

print_apim_preview <- function(df) {
  preview_cols <- c(
    "userID",
    "coupleID",
    "member",
    "gender",
    "is_male",
    "is_female",
    "satisfaction",
    "communication_actor_gmc",
    "communication_partner_gmc"
  )

  df |>
  dplyr::select(dplyr::any_of(preview_cols)) |>
  head() |>
  print_df()
}

options(
  brms.backend = 'cmdstan',
  brms.file_refit = 'on_change'
)

```

1 Distinguishable Dyads

2 Distinguishable Dyads - Cross-Sectional APIM

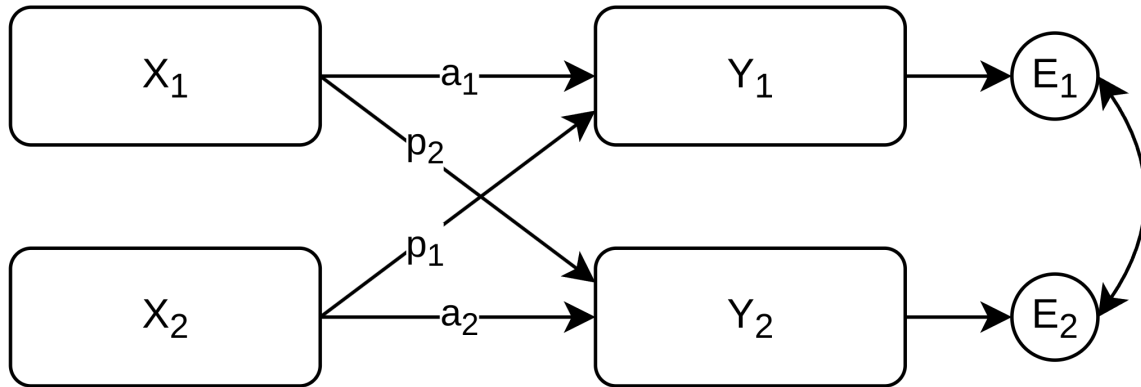
2.1 Distinguishable Dyads - Data: Simulated Dyads

```
print_df(head(df))
```

userID	coupleID	gender	comm	satisfaction
1 1	1	1	4.851107	4.841332

1 2	1	2	6.004533	5.577165
2 1	2	1	5.881321	7.248741
2 2	2	2	6.433723	6.960293
3 1	3	1	4.283971	6.928699
3 2	3	2	5.516060	6.077688

2.2 Distinguishable Dyads - Cross-Sectional APIM



(e.g., Kenny et al., 2006; Kenny & Cook, 1999; Kenny & Kashy, 2011)

2.3 Distinguishable Dyads - Preparing Data

```

df_apim <- df %>%
  # Reshaping to Actor-Partner format (4-field)
  reshape_dyadic_data(
    person_id = 'userID',
    dyad_id = 'coupleID',
    vars_to_reshape = 'communication'
  ) %>%
  mutate(
    # Optional: grand-mean centering
    communication_actor_gmc = communication_actor - mean(communication_actor),
    communication_partner_gmc = communication_partner - mean(communication_partner),

    # Create Dummy-Variables for male and female
    is_female = ifelse(gender == 1, 1, 0),
    is_male = ifelse(gender == 2, 1, 0),
    member = ifelse(is_female == 1, 1, 2)
  )

print_apim_preview(df_apim)

```

2.4 Distinguishable Dyads - Preparing Data

userID	coupleID	member	gender	is male	is female	satisfaction	comm actor	comm partner
1 1	1	1	1	0	1	4.841332	-0.1553604	0.9980657
1 2	1	2	2	1	0	5.577165	0.9980657	-0.1553604
2 1	2	1	1	0	1	7.248741	0.8748537	1.4272563
2 2	2	2	2	1	0	6.960293	1.4272563	0.8748537
3 1	3	1	1	0	1	6.928699	-0.7224960	0.5095933
3 2	3	2	2	1	0	6.077688	0.5095933	-0.7224960

2.5 Distinguishable Dyads - Fitting the Model in glmmTMB

- glmmTMB models residual variance/dispersion via `dispformula`, but residual covariance is handled through random-effect covariance structures.
- For Gaussian mixed models, `dispformula = ~ 0` fixes the residual variance to a tiny value. This pushes the Gaussian residual variance into the random-effect block.
- In this Gaussian dyadic model, the `us(0 + is_male + is_female | coupleID)` block therefore represents the 2x2 residual variance-covariance structure between partners.
- For non-Gaussian models, this is not an ordinary residual covariance on the response scale. The same random-effect trick adds latent Gaussian heterogeneity/correlation on the linear predictor scale.
- Do **not** use `dispformula = ~ 0` as a general non-Gaussian residual-covariance trick. For families with an estimable dispersion parameter, `dispformula` models that family-specific parameter via a log link; for families without one, such as Poisson or binomial, it is ignored. Partner-specific dispersion, e.g. `dispformula = ~ 0 + is_male + is_female`, is only appropriate when the family has a dispersion parameter and this is substantively intended.

2.6 Distinguishable Dyads - Fitting the Model in glmmTMB

```
1 model_dist_apim_glmmTMB <- glmmTMB(  
2   satisfaction ~  
3  
4   # Remove global intercept, introduce male and female intercepts.  
5   0 + is_male + is_female +  
6  
7   # Actor effect for male and female  
8   communication_actor_gmc:is_male +  
9   communication_actor_gmc:is_female +  
10  
11  # Partner effect for male and female  
12  communication_partner_gmc:is_male +  
13  communication_partner_gmc:is_female +  
14  
15  # Modelling residual non-independence with an unstructured 2x2 covariance.  
16  us(0 + is_male + is_female | coupleID)  
17  
18  # Gaussian response:  
19  # dispformula = ~ 0 fixes residual variance to a tiny value,  
20  # pushing the Gaussian residual variance into this random-effect block.  
21  , dispformula = ~ 0  
22  
23  # Non-Gaussian responses:  
24  # dispformula is family-specific dispersion, or ignored for families  
25  # without a dispersion parameter. Do not use ~0 as a residual trick.  
26  # If dispersion exists and should differ by partner, use e.g.:  
27  # dispformula = ~ 0 + is_male + is_female  
28  
29  , data = df_apim  
30  , family = gaussian  
31  )
```

2.7 Distinguishable Dyads - Fitting the Model in glmmTMB

Note that ‘unstructured’ is the default in glmmTMB, so instead of

```
us(0 + is_male + is_female | coupleID)
```

we could also remove `us` and simply write

```
(0 + is_male + is_female | coupleID).
```

unstructured means, that both `is_male` and `is_female` both get their own residual variance, and a correlation is freely estimated between the two. If there were more than two members (e.g., triads), unstructured would estimate a different correlation for each pair.

An unstructured matrix for just two persons is written as:

!TODO: insert matrix.

2.8 Distinguishable Dyads - Fitting the Model in glmmTMB

```
Family: gaussian ( identity )
Formula:      satisfaction ~ 0 + is_male + is_female + communication_actor_gmc:is_male +
      communication_actor_gmc:is_female + communication_partner_gmc:is_male +
      communication_partner_gmc:is_female + us(0 + is_male + is_female |      coupleID)
Dispersion:      ~0
Data: df_apim

      AIC      BIC    logLik -2*log(L)  df.resid
3776.7    3821.7   -1879.3    3758.7     1091

Random effects:

Conditional model:
Groups   Name      Variance Std.Dev. Corr
coupleID is_male    2.559    1.600
          is_female 1.682    1.297    0.51
Number of obs: 1100, groups:  coupleID, 550

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
is_male      4.60635    0.06821   67.54 < 2e-16 ***
is_female    5.61086    0.05530  101.47 < 2e-16 ***
is_male:communication_actor_gmc 1.84363    0.04931   37.39 < 2e-16 ***
is_female:communication_actor_gmc 1.59247    0.04112   38.72 < 2e-16 ***
is_male:communication_partner_gmc 0.16765    0.05072    3.31 0.000949 ***
is_female:communication_partner_gmc 0.31084    0.03997    7.78 7.49e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2.9 glmmTMB: Current limitation

Usually, simulation-based tools like DHARMa (Hartig, 2026) are great for residual diagnostics in complex (generalized) MLMs.

However, when we remove the native residuals `dispformula = ~ 0` the residual is, in practice, set to a tiny constant. This confuses model diagnostic packages!

Therefore, residuals need to be manually simulated and compared.

2.10 Distinguishable Dyads - Fitting the Model in brms

In brms, we can use brms-provided residual covariance structures directly. We do not have to set residuals internally to zero.

```

1 formula_dist_apim_b <- bf(
2   satisfaction ~
3
4   # Remove global intercept, introduce male and female intercepts.
5   0 + is_male + is_female +
6
7   # Actor effect for male and female
8   communication_actor_gmc:is_male +
9   communication_actor_gmc:is_female +
10
11  # Partner effect for male and female
12  communication_partner_gmc:is_male +
13  communication_partner_gmc:is_female +
14
15  # Modelling residual correlation. The member variable indexes the two
16  # within-couple residual positions used by the unstructured correlation.
17  unstr(time = member, gr = coupleID),
18
19  # Allowing two distinct residual variances for males vs females:
20  sigma = ~ 0 + is_male + is_female
21 )
22
23
24 priors_dist_apim_b <- c(
25   # prior(normal(2, 3), class = "Intercept"),
26   prior(normal(0, 5), class = "b"),
27   prior(student_t(3, 0, 1.5), class = "b", dpar = "sigma"),
28   prior(lkj(2), class = "cortime")
29 )
30
31 model_dist_apim_b <- brm(
32   formula = formula_dist_apim_b,
33   data = df_apim,
34   family = gaussian(link = identity),
35   prior = priors_dist_apim_b,
36   chains = 4,
37   cores = 4,
38   iter = 2000,
39   warmup = 1000,
40   seed = 123,
41   file = file.path('brms_cache', 'example1_dist_apim_b') # Cache the model
42 )

```

2.11 Distinguishable Dyads - Fitting the Model in brms

```

Family: gaussian
Links: mu = identity; sigma = log
Formula: satisfaction ~ 0 + is_male + is_female + communication_actor_gmc:is_male +
↪ communication_actor_gmc:is_female + communication_partner_gmc:is_male +
↪ communication_partner_gmc:is_female + unstr(time = member, gr = coupleID)
sigma ~ 0 + is_male + is_female
Data: df_apim (Number of observations: 1100)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

```

Correlation Structures:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
cortime(1,2)	0.51	0.03	0.44	0.57	1.00	4930	2949

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
is_male	4.61	0.07	4.48	4.73	1.00	4350	2996
is_female	5.61	0.05	5.50	5.71	1.00	4166	3096
is_male:communication_actor_gmc	1.84	0.05	1.75	1.94	1.00	3930	3062
is_female:communication_actor_gmc	1.59	0.04	1.51	1.68	1.00	3649	2492
is_male:communication_partner_gmc	0.17	0.05	0.07	0.27	1.00	3574	2949
is_female:communication_partner_gmc	0.31	0.04	0.23	0.39	1.00	3976	3169
sigma_is_male	0.47	0.03	0.41	0.53	1.00	5196	2993
sigma_is_female	0.26	0.03	0.20	0.32	1.00	5500	3150

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat = 1`).

2.12 Distinguishable Dyads - Fitting the Model in brms

For families that have no residuals, this does not work and we need to model the non-independence in the random effects:

```

1 formula_dist_apim_c <- bf(
2   as.integer(satisfaction + 6) ~
3
4   # Remove global intercept, introduce male and female intercepts.
5   0 + is_male + is_female +
6
7   # Actor effect for male and female
8   communication_actor_gmc:is_male +
9   communication_actor_gmc:is_female +
10
11  # Partner effect for male and female
12  communication_partner_gmc:is_male +
13  communication_partner_gmc:is_female +
14
15  # Modelling residual correlation as random effects, because we cannot specify
16  # a sigma parameter, as this distribution has no residuals.
17
18  (0 + is_male + is_female | coupleID)
19
20  # sigma ~ 0 + is_male + is_female
21 )
22
23 priors_dist_apim_c <- c(
24   prior(normal(0, 5), class = "b"),
25   prior(student_t(3, 0, 1.5), class = "sd"),
26   prior(lkj(2), class = "cor")
27 )
28
29 model_dist_apim_c <- brm(
30   formula = formula_dist_apim_c,
31   data = df_apim,
32   family = poisson(),
33   prior = priors_dist_apim_c,
34   chains = 4,
35   cores = 4,
36   iter = 2000,
37   warmup = 1000,
38   seed = 123,
39   file = file.path('brms_cache', 'example1_dist_apim_c') # Cache the model
40 )

```

2.13 Distinguishable Dyads - Fitting the Model in brms

```

Family: poisson
Links: mu = log
Formula: as.integer(satisfaction + 6) ~ 0 + is_male + is_female + communication_actor_gmc:is_male +
↪ communication_actor_gmc:is_female + communication_partner_gmc:is_male +
↪ communication_partner_gmc:is_female + (0 + is_male + is_female | coupleID)
Data: df_apim (Number of observations: 1100)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000

Multilevel Hyperparameters:
~coupleID (Number of levels: 550)

```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(is_male)	0.01	0.01	0.00	0.04	1.00	3258	1713

```
sd(is_female)      0.01      0.01      0.00      0.03 1.00      2400      1244
cor(is_male,is_female) 0.02      0.46     -0.83      0.84 1.00      5097      2128
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
is_male	2.27	0.01	2.25	2.30	1.00	5852	2853
is_female	2.38	0.01	2.35	2.40	1.00	6236	2817
is_male:communication_actor_gmc	0.18	0.01	0.16	0.20	1.00	4083	2735
is_female:communication_actor_gmc	0.14	0.01	0.13	0.16	1.00	4607	2949
is_male:communication_partner_gmc	0.02	0.01	-0.00	0.04	1.00	5095	3065
is_female:communication_partner_gmc	0.03	0.01	0.01	0.05	1.00	4312	3116

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

2.14 Distinguishable Dyads - Fitting the Model in glmmTMB

```
model_glmmTMB_dist_apim_c <- glmmTMB(
  as.integer(satisfaction + 6) ~

  # Remove global intercept, introduce male and female intercepts.
  0 + is_male + is_female +

  # Actor effect for male and female
  communication_actor_gmc:is_male +
  communication_actor_gmc:is_female +

  # Partner effect for male and female
  communication_partner_gmc:is_male +
  communication_partner_gmc:is_female +

  # Modelling residual correlation as random effects, because we cannot specify
  # a sigma parameter, as this distribution has no residuals.

  (0 + is_male + is_female | coupleID)
, data = df_apim
, family = poisson()
)
```

2.15 Distinguishable Dyads - Fitting the Model in glmmTMB

```
Family: poisson ( log )
Formula:          as.integer(satisfaction + 6) ~ 0 + is_male + is_female + communication_actor_gmc:is_male
~ +
  communication_actor_gmc:is_female + communication_partner_gmc:is_male +
  communication_partner_gmc:is_female + (0 + is_male + is_female | coupleID)
Data: df_apim

      AIC      BIC    logLik -2*log(L)  df.resid
      NA      NA      NA      NA      1091

Random effects:

Conditional model:
Groups   Name      Variance Std.Dev.  Corr
coupleID is_male    6.697e-11 8.184e-06
          is_female  7.510e-11 8.666e-06 1.00
Number of obs: 1100, groups: coupleID, 550

Conditional model:

      Estimate Std. Error z value Pr(>|z|)
is_male    2.274016   0.013933 163.21 < 2e-16 ***
is_female    2.379174   0.013143 181.02 < 2e-16 ***
is_male:communication_actor_gmc 0.180139   0.009598  18.77 < 2e-16 ***
```

```
is_female:communication_actor_gmc 0.143973 0.009580 15.03 < 2e-16 ***
is_male:communication_partner_gmc 0.015476 0.010003 1.55 0.12183
is_female:communication_partner_gmc 0.028344 0.009180 3.09 0.00202 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2.16 Distinguishable Dyads - Check Model Convergence and Fit

Check Rhats and Effective Sample Sizes (ESS_tail and ESS_bulk) directly from the brms summary. Additionally you can (using Model B as an example):

```
rstan::check_hmc_diagnostics(model_dist_apim_b$fit)
```

Divergences:

0 of 4000 iterations ended with a divergence.

Tree depth:

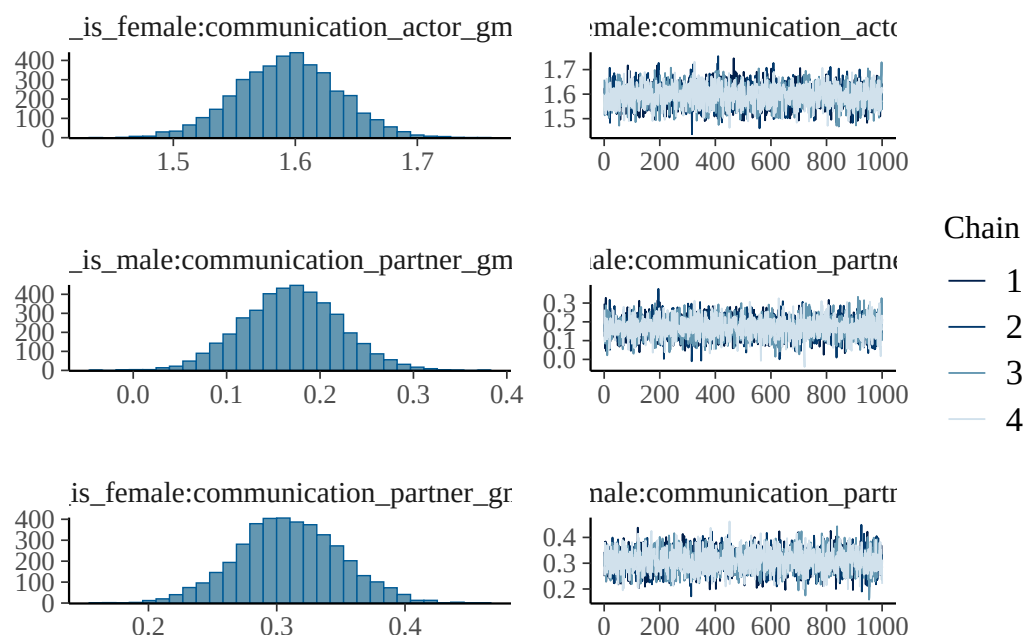
0 of 4000 iterations saturated the maximum tree depth of 10.

Energy:

E-BFMI indicated no pathological behavior.

2.17 Distinguishable Dyads - Check Model Convergence and Fit

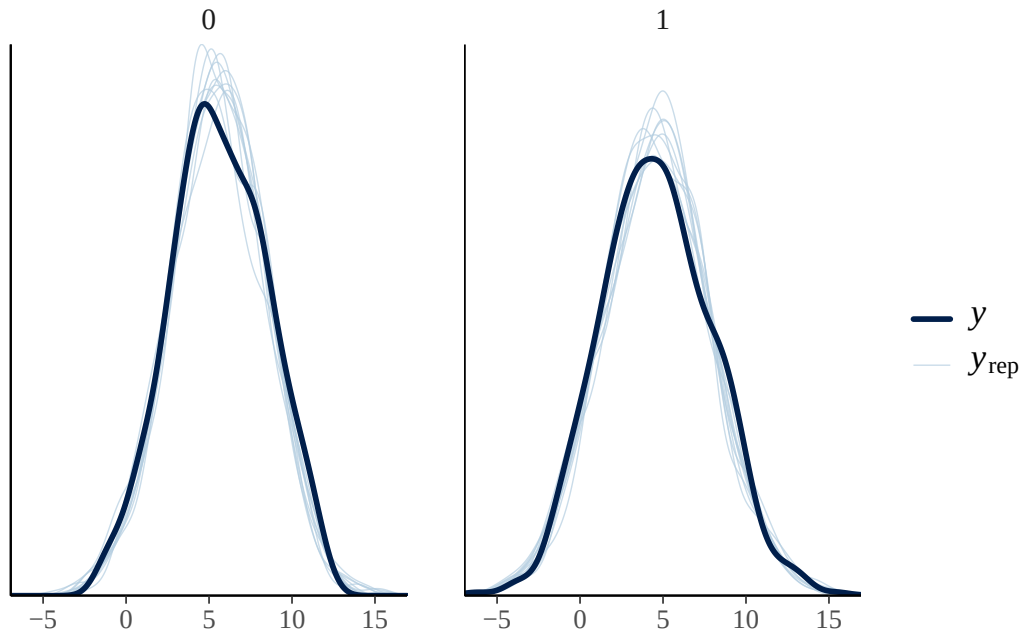
```
plot(model_dist_apim_b, ask = FALSE)
```



2.18 Distinguishable Dyads - Check Model Convergence and Fit

```
pp_check(model_dist_apim_b, 'dens_overlay_grouped', group = 'is_male')
```

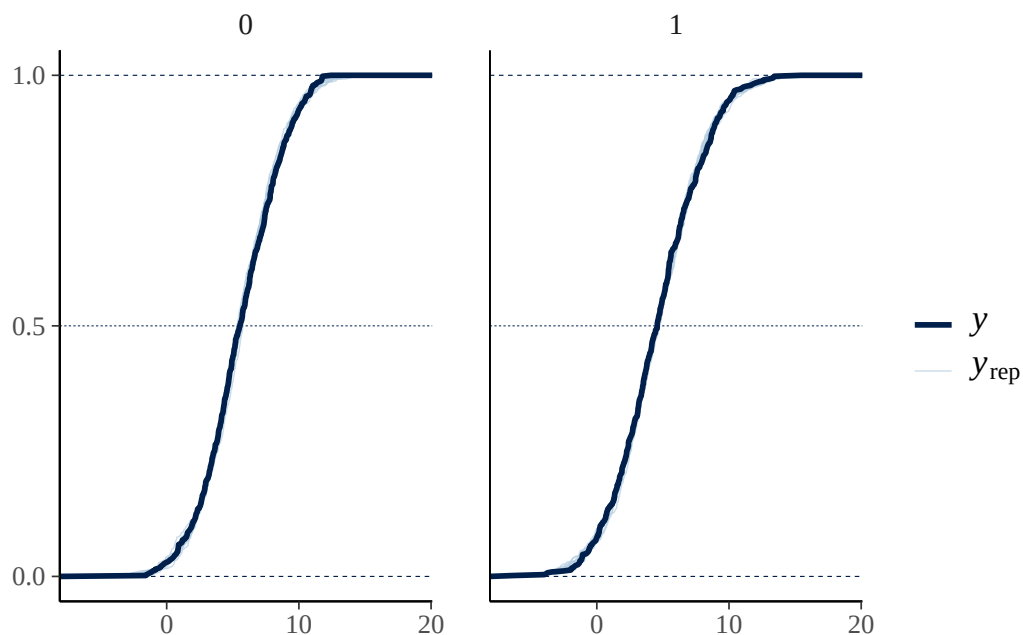
Using 10 posterior draws for ppc type 'dens_overlay_grouped' by default.



2.19 Distinguishable Dyads - Check Model Convergence and Fit

```
pp_check(model_dist_apim_b, 'ecdf_overlay_grouped', group = 'is_male')
```

Using 10 posterior draws for ppc type 'ecdf_overlay_grouped' by default.

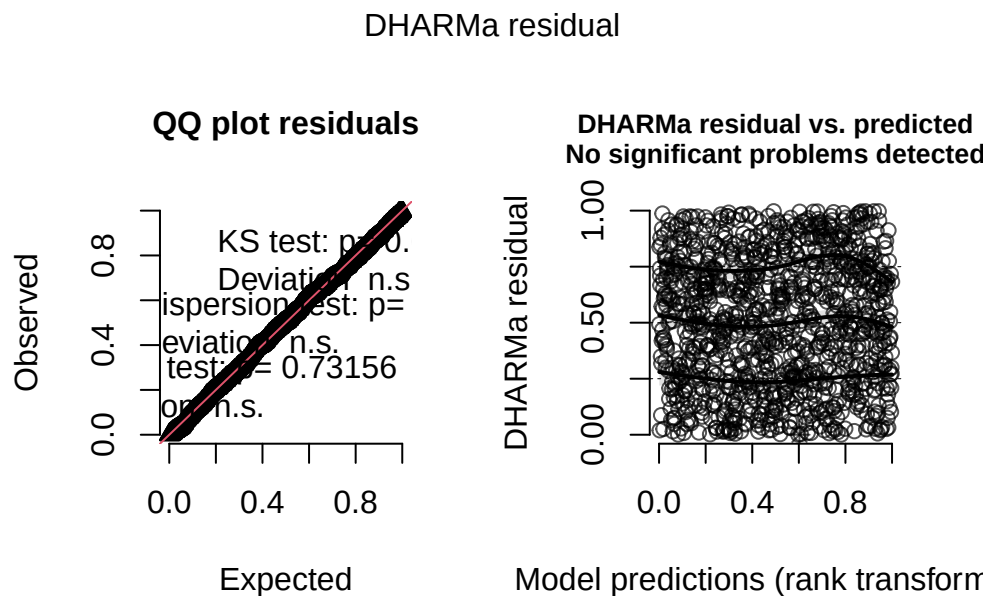


2.20 Distinguishable Dyads - Check Model Convergence and Fit

```
pp_check(model_dist_apim_b, type = "loo_pit_overlay")
```

2.21 Distinguishable Dyads - Check Model Convergence and Fit

```
# Custom function to make DHARMA work with brms (see file 'Functions')
DHARMA.check_brms(model_dist_apim_b)
```



2.22 Distinguishable Dyads - Check Model Convergence and Fit

In case of non-convergence, bad residuals or low predictive accuracy, the model is likely misspecified.

You can try different families that match your data by using the `family` argument. For instance, you can easily conduct an ordinal regression by simply setting `family = cumulative()` and adjusting your priors. For dichotomous outcomes you can use `family = bernoulli()`. `brms` supports a variety of families and link functions for these families. In general, you should use the default link function (check `?brms::family.brmsfit()`).

Note that for some of the later models in this tutorial, some ESS values are too low. In this case, usually we would increase iterations (and warmup), and/or `adapt_delta` (see `brms` documentation).

2.23 Excursion: Generalized Dyadic Models

For Gaussian outcomes, residual covariance has a direct response-scale interpretation:

- partner-specific residual variances
- partner residual correlations
- same-day residual covariance in ILD data

For non-Gaussian outcomes, the same syntax usually does **not** estimate an ordinary residual covariance.

Instead, random-effect covariance terms represent latent Gaussian heterogeneity on the link scale.

2.24 Excursion: Count vs Binary Outcomes

For count outcomes, latent partner covariance is often useful and can be estimable:

- Poisson or negative-binomial models can include dyad-level latent covariance terms
- estimates live on the log-link scale
- covariance parameters need diagnostics and sensitivity checks

For dichotomous outcomes, the same idea is much harder:

- each partner contributes only 0 or 1 per occasion
- many latent variance-correlation combinations can imply similar pair probabilities
- same-day binary covariance is especially weakly identified

2.25 Excursion: Practical Warning

Binary dyadic covariance models can converge mechanically while still being unreliable.

Watch for:

- variance estimates near zero or extremely large
- correlations near -1 or 1
- low ESS, divergent transitions, or strong prior sensitivity
- different software giving very different covariance estimates

2.26 Excursion: Practical Strategy

Practical default:

- estimate simpler structures first
- use informative priors if fitting latent binary covariance in brms
- treat same-day binary partner covariance as a specialized model, not a routine extension
- if the main goal is fixed effects, consider simpler random effects or robust/clustered inference

2.27 Distinguishable Dyads - Results

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
Fixed Effects					
is male	4.61	[4.48, 4.73]	1.000	4350	2996
is female	5.61	[5.50, 5.71]	1.000	4166	3096
is male: comm actor	1.84	[1.75, 1.94]	1.000	3930	3062
is female: comm actor	1.59	[1.51, 1.68]	1.000	3649	2492
is male: comm partner	0.17	[0.07, 0.27]	1.002	3574	2949
is female: comm partner	0.31	[0.23, 0.39]	1.001	3976	3169
Residual Structure					
cortime(1, 2)	0.51	[0.44, 0.57]	1.000	4930	2949

sigma is male	1.60	[1.51, 1.71]	1.000	5196	2993
sigma is female	1.30	[1.23, 1.38]	1.001	5500	3150

NOTE: brms estimates sigma on the log scale. In the tables, sigmas are exponentiated to show residual SDs on the response scale, like glmmTMB output.

3 Distinguishable Dyads

4 Distinguishable Dyads - Intensive Longitudinal APIM (L-APIM)

4.1 Distinguishable Dyads - L-APIM - Simulated Dataset

userID	coupleID	couple day	day	day c	gender	member	is male	is female	closeness	support actor	support partner
31 1	31	31.0	0	-27	1	1	0	1	5.47	6.02	7.09
31 1	31	31.1	1	-26	1	1	0	1	7.69	6.13	6.95
31 1	31	31.2	2	-25	1	1	0	1	5.83	6.1	6.34
31 1	31	31.3	3	-24	1	1	0	1	6.55	6.99	5.39
...	...	...	...	...	...	...	...	...	...	...	...
31 2	31	31.0	0	-27	2	2	1	0	5.19	7.09	6.02
31 2	31	31.1	1	-26	2	2	1	0	6.96	6.95	6.13
31 2	31	31.2	2	-25	2	2	1	0	6.9	6.34	6.1
31 2	31	31.3	3	-24	2	2	1	0	7.59	5.39	6.99

4.2 Distinguishable Dyads - L-APIM - Centering

With repeated measures we need to center variables in order to not conflate levels of analysis:

- Between-Person: The person-mean in relation to the grand mean
- Within-Person: Daily fluctuations of an individual from their person-mean

Although the data contain couple, person, and day structure, the APIM is written at the person-day level after decomposing predictors into between-person and within-person components. Dyad-level dependence is still modelled explicitly through partner-specific random effects and residual covariance structures.

4.3 Distinguishable Dyads - L-APIM - Centering

```
df_long_apim <- bmlm::isolate(
  df_long_apim,
  by = 'userID', # NOT coupleID
  value = c('provided_support_actor', 'provided_support_partner'),
  which = 'both'
) %>%
# renaming to avoid confusion with between- and within COUPLE centred DIM variables
rename(
  provided_support_actor_cwp = provided_support_actor_cw,
  provided_support_partner_cwp = provided_support_partner_cw,
  provided_support_actor_cbp = provided_support_actor_cb,
  provided_support_partner_cbp = provided_support_partner_cb,
) %>%
# Centering time
mutate(
  diaryday_c = scale(diaryday, center = TRUE, scale = FALSE)
)
```

4.4 Distinguishable Dyads - L-APIM - Model nlme

nlme model of this data for reference (simplified random effects structure to achieve convergence).

This package is old, but still often used and cited. So here, we briefly show it to demonstrate that we can achieve the same model with much more modern and powerful methods.

```
library(nlme)

model1 <- lme(
  fixed =
    # Intercepts
    closeness ~ 0 + is_male + is_female +

    # Time Slopes
    is_male:diaryday_c +
    is_female:diaryday_c +

    # Between-Person APIM
    is_male:provided_support_actor_cbp +
    is_male:provided_support_partner_cbp +

    is_female:provided_support_actor_cbp +
    is_female:provided_support_partner_cbp +

    # Within-Person APIM
    is_male:provided_support_actor_cwp +
    is_male:provided_support_partner_cwp +

    is_female:provided_support_actor_cwp +
    is_female:provided_support_partner_cwp,

  random =
    ~ 0 + is_male + is_female

    + is_male:diaryday_c + is_female:diaryday_c

  # in nlme we need to remove many random slopes to achieve convergence
  # alternatively, we could try to remove correlations between the slopes.
  #+ is_male:provided_support_actor_cwp
  #+ is_male:provided_support_partner_cwp
  #+ is_female:provided_support_actor_cwp
  #+ is_female:provided_support_partner_cwp
  | coupleID,

  weights = varIdent(form = ~ 1 | gender), # heterogeneous residual variances
  corr = corCompSymm(form = ~ 1 | coupleID/diaryday), # compound symmetry (partner's values on each day)

  data = df_long_apim,
  control = list(maxIter=1000)
)

summary(model1)
```

4.5 Distinguishable Dyads - L-APIM - Model glmmTMB

```
1 model_dist_apim_long_glmmTMB <- glmmTMB(
2   closeness ~ 0 + is_male + is_female +
3
4   # Time Slopes
5   is_male:diaryday_c +
6   is_female:diaryday_c +
7
8   # Between-Person APIM
9   is_male:provided_support_actor_cbp +
10  is_male:provided_support_partner_cbp +
```



```

11
12   is_female:provided_support_actor_cbp +
13   is_female:provided_support_partner_cbp +
14
15   # Within-Person APIM
16   is_male:provided_support_actor_cwp +
17   is_male:provided_support_partner_cwp +
18
19   is_female:provided_support_actor_cwp +
20   is_female:provided_support_partner_cwp +
21
22   # Accounting for non-independence between partners' means and trajectories
23   # and effect sensitivities via random effects:
24   (0 + is_male + is_female
25     + is_male:diaryday_c + is_female:diaryday_c
26     | coupleID) +
27
28   # Accounting for daily non-independence
29   us(0 + is_male + is_female | coupleID:diaryday)
30
31   # heteroscedastic residuals
32   , dispformula = ~ 0
33   , data = df_long_apim
34 )

```

4.6 Distinguishable Dyads - L-APIM - Model glmmTMB

```

Family: gaussian ( identity )
Formula:      closeness ~ 0 + is_male + is_female + is_male:diaryday_c + is_female:diaryday_c +
  is_male:provided_support_actor_cbp + is_male:provided_support_partner_cbp +
  is_female:provided_support_actor_cbp + is_female:provided_support_partner_cbp +
  is_male:provided_support_actor_cwp + is_male:provided_support_partner_cwp +
  is_female:provided_support_actor_cwp + is_female:provided_support_partner_cwp +
  (0 + is_male + is_female + is_male:diaryday_c + is_female:diaryday_c |
    coupleID) + us(0 + is_male + is_female | coupleID:diaryday)
Dispersion:      ~0
Data: df_long_apim

      AIC      BIC    logLik -2*log(L)  df.resid
30001.8  30184.4 -14975.9  29951.8    10975

Random effects:

Conditional model:
Groups      Name      Variance Std.Dev. Corr
coupleID    is_male    0.6814567 0.82550
            is_female  0.7929922 0.89050 0.19
            is_male:diaryday_c 0.0002637 0.01624 0.55 0.03
            is_female:diaryday_c 0.0002571 0.01603 0.21 0.54 -0.01
coupleID:diaryday is_male 1.2583655 1.12177
            is_female  0.5206624 0.72157 0.04
Number of obs: 11000, groups:  coupleID, 100; coupleID:diaryday, 5500

Conditional model:

```

	Estimate	Std. Error	z value	Pr(> z)
is_male	4.486339	0.085965	52.19	< 2e-16 ***
is_female	5.736714	0.091763	62.52	< 2e-16 ***
is_male:diaryday_c	-0.003259	0.001883	-1.73	0.08350 .
is_female:diaryday_c	0.010605	0.001717	6.18	6.48e-10 ***
is_male:provided_support_actor_cbp	1.127989	0.101927	11.07	< 2e-16 ***
is_male:provided_support_partner_cbp	0.298916	0.098822	3.02	0.00249 **
is_female:provided_support_actor_cbp	1.380810	0.105605	13.08	< 2e-16 ***
is_female:provided_support_partner_cbp	0.587224	0.108925	5.39	7.00e-08 ***
is_male:provided_support_actor_cwp	0.184466	0.020564	8.97	< 2e-16 ***
is_male:provided_support_partner_cwp	0.124546	0.020571	6.05	1.41e-09 ***
is_female:provided_support_actor_cwp	0.419160	0.013277	31.57	< 2e-16 ***
is_female:provided_support_partner_cwp	0.170136	0.013270	12.82	< 2e-16 ***

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

4.7 Comparing outputs between nlme and glmmTMB

Let's compare the random effects outputs to see if the models are truly equivalent.

Residual variance in `nlme` must first be constructed from the output by taking the residual SD and multiplying it by the parameter estimate under the variance function for each partner.

nlme:

	StdDev	Corr			
is_male	0.83645552		is_mal	is_fml	is_m:_
is_female	0.90199634	0.189			
is_male:diaryday_c	0.01634796	0.544	0.033		
is_female:diaryday_c	0.01612665	0.204	0.534	-0.012	
Residual	0.72170702				

Correlation Structure: Compound symmetry

Formula: ~1 | coupleID/diaryday

Parameter estimate(s):

Rho

0.04094751

Variance function:

Structure: Different standard deviations per stratum

Formula: ~1 | gender

Parameter estimates:

1	2
---	---

1.000000	1.554608
----------	----------

Manually computed by multiplying the residual SD by the multipliers:

Residual SDs by partner:

Partner 1: 0.721707

Partner 2: 1.121972

glmmTMB:

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.	Corr			
coupleID	is_male	0.6814498	0.82550				
	is_female	0.7930407	0.89053	0.19			
	is_male:diaryday_c	0.0002637	0.01624	0.55	0.03		
	is_female:diaryday_c	0.0002571	0.01603	0.21	0.54	-0.01	
coupleID:diaryday	is_male	1.2583552	1.12176				
	is_female	0.5206608	0.72157	0.04			

4.8 Distinguishable Dyads - L-APIM - Model brms

Equivalent brms model.

In contrast to glmmTMB, brms provides native variance-covariance structures to model residuals directly. Therefore, we do NOT set the residuals to zero!

```
1 formula_dist_apim_long_simple <- bf(
2   closeness ~ 0 + is_male + is_female +
3
4   # Time Slopes
5   is_male:diaryday_c +
6   is_female:diaryday_c +
7
8   # Between-Person APIM
9   is_male:provided_support_actor_cbp +
10  is_male:provided_support_partner_cbp +
11
12  is_female:provided_support_actor_cbp +
13  is_female:provided_support_partner_cbp +
14
15  # Within-Person APIM
16  is_male:provided_support_actor_cwp +
17  is_male:provided_support_partner_cwp +
18
19  is_female:provided_support_actor_cwp +
20  is_female:provided_support_partner_cwp +
21
22  # Accounting for non-independence between partners' means and trajectories
23  # and effect sensitivities via random effects:
24  (0 + is_male + is_female
25   + is_male:diaryday_c + is_female:diaryday_c
26   | coupleID) +
27
28  # Accounting for same-day residual correlation. The member variable indexes
29  # the two partner residual positions within each couple-day.
30  unstr(time = member, gr = couple_day)
31
32  # heteroscedastic residuals
33  , sigma ~ 0 + is_male + is_female
34 )
35
36 # brms' defaults for group-level SDs and correlations are sensible weak defaults
37 # (half Student-t priors for SDs and LKJ priors for correlation matrices). Here we
38 # set weakly regularizing priors explicitly so the modelling assumptions are visible.
39
40 priors_dist_apim_long_simple <- c(
41   prior(normal(4, 2), class = "b", coef = "is_male"), # male intercept
42   prior(normal(4, 2), class = "b", coef = "is_female"), # female intercept
43   prior(normal(0, 2), class = "b"), # other beta coefficients
44   prior(exponential(1), class = "sd"),
45   prior(lkj(2), class = "cor"),
46   prior(normal(0, 0.7), class = "b", dpar = "sigma"),
47   prior(lkj(2), class = "cortime")
48 )
49
50 model_dist_apim_long_simple <- brm(
51   formula = formula_dist_apim_long_simple,
52   data = df_long_apim,
53   family = gaussian(link = identity), # student() is also often a nice robust option.
54   prior = priors_dist_apim_long_simple,
55   chains = 4,
56   cores = 4,
57   iter = 2000,
58   warmup = 1000,
59   seed = 123,
60   file = file.path('brms_cache', 'example1_dist_apim_long_simpel') # Cache the model
61 )
```

4.9 Distinguishable Dyads - L-APIM - Results (scrollable)

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
Fixed Effects					
is male	4.49	[4.32, 4.66]	1.011	433	602
is female	5.74	[5.57, 5.91]	1.016	368	703
is male: day c	0.00	[-0.01, 0.00]	1.001	1650	2331
is female: day c	0.01	[0.01, 0.01]	1.001	960	1754
is male: support actor BP	1.13	[0.92, 1.33]	1.001	609	1163
is male: support partner BP	0.31	[0.11, 0.50]	1.004	506	1070
is female: support actor BP	1.38	[1.16, 1.60]	1.016	312	842
is female: support partner BP	0.59	[0.38, 0.81]	1.002	458	1046
is male: support actor WP	0.18	[0.14, 0.23]	1.003	7382	2557
is male: support partner WP	0.12	[0.08, 0.17]	1.001	9272	2694
is female: support actor WP	0.42	[0.39, 0.44]	1.002	7263	2967
is female: support partner WP	0.17	[0.15, 0.20]	1.000	8513	3389
Random Effects					
sd(is male)	0.84	[0.74, 0.97]	1.003	698	1502
sd(is female)	0.91	[0.80, 1.06]	1.002	567	1115
sd(is male: day c)	0.02	[0.01, 0.02]	1.002	1741	2708
sd(is female: day c)	0.02	[0.01, 0.02]	1.003	1462	2124
cor(is male, is female)	0.18	[-0.02, 0.35]	1.006	417	560
cor(is male, is male: day c)	0.51	[0.32, 0.67]	1.001	1738	2782
cor(is female, is male: day c)	0.02	[-0.19, 0.23]	1.002	1590	2609
cor(is male, is female: day c)	0.19	[-0.01, 0.38]	1.002	994	1799
cor(is female, is female: day c)	0.51	[0.33, 0.64]	1.004	1365	2057
cor(is male: day c, is female: day c)	-0.01	[-0.25, 0.21]	1.002	873	2086
cortime(1, 2)	0.04	[0.02, 0.07]	1.001	8852	2962
Residual Structure					
sigma is male	1.12	[1.10, 1.14]	1.000	9812	2936
sigma is female	0.72	[0.71, 0.74]	1.000	9002	2896

Note: Sigma rows are shown on the response scale; they were exponentiated from brms' log-sigma parameterization.

4.10 Distinguishable Dyads - L-APIM - AR(1)?

In this model, same-day partner interdependence is handled by `unstr(time = member, gr = couple_day)`. In current univariate brms models, we cannot combine this residual correlation structure with `ar()` in the same formula.

glmmTMB currently also does not allow partner-interdependence across time, but I have a fork of glmmTMB with added support and am currently working of integrating this with upstream glmmTMB.

4.11 Excursion: Dynamic Workaround

A dynamic MLM can account for temporal dependence by including lagged within-person centered outcomes as predictors (include both actor inertia and partner spillover).

But:

- The APIM coefficients become effects conditional on prior closeness, not total within-person associations.
- Lagged-outcome models can also suffer from dynamic-panel bias, often called Nickell bias (Nickell, 1981), because the lagged outcome is related to person/couple effects and prior residuals. This is especially problematic with short time series.
 - One solution for the Nickell bias is to use multilevel DSEM through MPLUS.

4.12 Maximal Model Specification (non-dynamic/no AR(1))

We add all random slopes and their correlations. This is neither converging in `nlme` nor `glmmTMB`.

```
1 formula_dist_apim_long_complex <- bf(
2   closeness ~
3
4   # Intercepts
5   0 + is_male + is_female +
6
7   # Time Slopes
8   is_male:diaryday_c +
9   is_female:diaryday_c +
10
11  # Between-Person APIM
12  is_male:provided_support_actor_cbp +
13  is_male:provided_support_partner_cbp +
14
15  is_female:provided_support_actor_cbp +
16  is_female:provided_support_partner_cbp +
17
18  # Within-Person APIM
19  is_male:provided_support_actor_cwp +
20  is_male:provided_support_partner_cwp +
21
22  is_female:provided_support_actor_cwp +
23  is_female:provided_support_partner_cwp +
24
25  # Accounting for non-independence between partners' means and trajectories
26  # and effect sensitivities via random effects:
27  (0 + is_male + is_female +
28
29   is_male:diaryday_c + is_female:diaryday_c +
30
31   is_male:provided_support_actor_cwp +
32   is_male:provided_support_partner_cwp +
33   is_female:provided_support_actor_cwp +
34   is_female:provided_support_partner_cwp
35  | coupleID ) +
36
37  # Accounting for daily non-independence
38  unstr(time = member, gr = couple_day),
39
40  # No more AR1. You could also test MA or ARMA.
41
42  sigma ~ 0 + is_male + is_female # heteroscedastic residuals
43 )
44
45 priors_dist_apim_long_complex <- c(
46   prior(normal(4, 2), class = "b", coef = "is_male"),
47   prior(normal(4, 2), class = "b", coef = "is_female"),
48   prior(normal(0, 2), class = "b"),
49   prior(exponential(1), class = "sd"),
50   prior(lkj(2), class = "cor"),
51   prior(normal(0, 0.7), class = "b", dpar = "sigma"),
52   prior(lkj(2), class = "cortime")
53 )
54
55
56 model_dist_apim_long_complex <- brm(
57   formula = formula_dist_apim_long_complex,
58   data = df_long_apim,
59   family = gaussian(link = identity),
60   prior = priors_dist_apim_long_complex,
61   chains = 4,
62   cores = 4,
63   iter = 2000,
64   warmup = 1000,
65   seed = 123,
66   file = file.path('brms_cache', 'example1_dist_apim_long_complex') # Cache the model
67 )
```

4.13 Maximal Model Specification: Results

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
Fixed Effects					
is male	4.48	[4.31, 4.65]	1.002	472	1126
is female	5.74	[5.54, 5.92]	1.010	207	602
is male: day c	0.00	[-0.01, 0.00]	1.001	1589	2542
is female: day c	0.01	[0.01, 0.01]	1.001	844	1709
is male: support actor	1.11	[0.90, 1.32]	1.015	437	811
BP					
is male: support	0.31	[0.12, 0.50]	1.013	497	1096
partner BP					
is female: support actor	1.38	[1.15, 1.60]	1.009	345	563
BP					
is female: support	0.59	[0.38, 0.80]	1.004	470	693
partner BP					
is male: support actor	0.18	[0.14, 0.23]	1.001	5396	3415
WP					
is male: support	0.13	[0.08, 0.17]	1.001	5133	3487
partner WP					
is female: support actor	0.42	[0.39, 0.45]	1.000	3981	2949
WP					
is female: support	0.17	[0.14, 0.20]	1.001	5813	2946
partner WP					
Random Effects					
sd(is male)	0.84	[0.73, 0.97]	1.009	623	1088
sd(is female)	0.91	[0.80, 1.06]	1.008	601	1324
sd(is male: day c)	0.02	[0.01, 0.02]	1.001	2150	2962
sd(is female: day c)	0.02	[0.01, 0.02]	1.002	1487	2367
sd(is male: support	0.09	[0.02, 0.16]	1.006	1030	935
actor WP)					
sd(is male: support	0.13	[0.03, 0.20]	1.010	662	448
partner WP)					
sd(is female: support	0.08	[0.02, 0.12]	1.008	919	802
actor WP)					
sd(is female: support	0.05	[0.00, 0.10]	1.003	764	1058
partner WP)					
cor(is male, is female)	0.16	[-0.04, 0.35]	1.005	301	657
cor(is male, is male:	0.48	[0.28, 0.64]	1.003	1505	2464
day c)					
cor(is female, is male:	0.01	[-0.20, 0.24]	1.002	1746	2332
day c)					
cor(is male, is female:	0.18	[-0.03, 0.37]	1.001	791	2042
day c)					
cor(is female, is female:	0.49	[0.32, 0.63]	1.001	1143	2201
day c)					
cor(is male: day c, is	-0.03	[-0.26, 0.20]	1.002	1117	2297
female: day c)					
cor(is male, is male:	0.37	[-0.06, 0.70]	1.002	4576	3139
support actor WP)					
cor(is female, is male:	0.19	[-0.22, 0.57]	1.000	4741	2951
support actor WP)					
cor(is male: day c, is	0.32	[-0.16, 0.68]	1.001	3912	1878
male: support actor WP)					
cor(is female: day c, is	0.27	[-0.17, 0.64]	1.000	4908	2510
male: support actor WP)					
cor(is male, is male:	0.06	[-0.27, 0.40]	1.001	4405	2894
support partner WP)					
cor(is female, is male:	0.11	[-0.24, 0.44]	1.001	4278	2751
support partner WP)					
cor(is male: day c, is	0.05	[-0.33, 0.42]	1.001	3650	2761
male: support partner					
WP)					
cor(is female: day c, is	0.12	[-0.24, 0.44]	1.000	3883	2641
male: support partner					
WP)					
cor(is male: support	0.25	[-0.29, 0.68]	1.003	862	1487
actor WP, is male:					
support partner WP)					
cor(is male, is female:	-0.06	[-0.42, 0.29]	1.001	3708	3083
support actor WP)					
cor(is female, is female:	0.20	[-0.17, 0.53]	1.003	3939	1795
support actor WP)					

(continued)

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
cor(is male: day c, is female: support actor WP)	-0.14	[-0.51, 0.25]	1.002	4113	3003
cor(is female: day c, is female: support actor WP)	-0.08	[-0.42, 0.28]	1.002	4693	2979
cor(is male: support actor WP, is female: support actor WP)	-0.16	[-0.62, 0.41]	1.002	1072	1594
cor(is male: support partner WP, is female: support actor WP)	-0.11	[-0.57, 0.38]	1.000	2068	2732
cor(is male, is female: support partner WP)	0.27	[-0.22, 0.64]	1.003	3771	2117
cor(is female, is female: support partner WP)	0.06	[-0.39, 0.50]	1.001	4733	1954
cor(is male: day c, is female: support partner WP)	0.11	[-0.36, 0.54]	1.000	4944	2757
cor(is female: day c, is female: support partner WP)	0.15	[-0.35, 0.57]	1.000	5159	2468
cor(is male: support actor WP, is female: support partner WP)	0.17	[-0.41, 0.65]	1.000	1967	2687
cor(is male: support partner WP, is female: support partner WP)	0.06	[-0.47, 0.56]	1.000	3064	2818
cor(is female: support actor WP, is female: support partner WP)	0.09	[-0.45, 0.60]	1.001	2596	2481
cortime(1, 2)	0.04	[0.01, 0.07]	1.002	8914	2967
Residual Structure					
sigma is male	1.11	[1.09, 1.14]	1.004	6856	3032
sigma is female	0.72	[0.70, 0.73]	1.000	6102	2899

4.14 Maximal Model Specification: Diagnostics

In complex models like this, we want to make sure we are not **overfitting**.

- Conduct regular model convergence checks (inspecting chains, Rhats, ESS, Multicollinearity, Residual diagnostics etc.). If these are bad (but good for a simpler model), it could be a sign of overfitting.
- Leave-one-out cross-validation is a powerful tool for investigating overfitting. It tests out of sample prediction and thus punishes overfitting.

4.15 Maximal Model Specification: Diagnostics

```
loo1 <- loo(model_dist_apim_long_simple)
loo2 <- loo(model_dist_apim_long_complex)

loo::pareto_k_table(loo1)
loo::pareto_k_table(loo2)
```

All Pareto k estimates are good ($k < 0.7$).

All Pareto k estimates are good ($k < 0.7$).

High Pareto-k values indicate influential observations and/or unreliable importance-sampling approximations. They are not automatically evidence of overfitting, but they should be checked; `loo(model, moment_match = TRUE)` is often recommended when `loo` flags problematic values.

The absolute ELPD values are not interpreted in isolation, but differences in ELPD can be used to compare models.

4.16 Maximal Model Specification: Diagnostics

```
a <- loo_compare(loo1, loo2)
print(a)
```

	elpd_diff	se_diff
model_dist_apim_long_complex	0.0	0.0
model_dist_apim_long_simple	-15.4	6.7

No obvious overfitting is detected here. The more complex model appears more predictive, but ELPD differences should be interpreted relative to their standard errors; rules such as 2x SE are rough heuristics, not hard decision thresholds.

If there was no obvious difference, we could follow our theory and the philosophy of the maximal random effects structure (Barr et al., 2013). Alternatively, we could follow the philosophy of parsimony, especially in the territory of such complex models.

(continued on next page)

4.17 Maximal Model Specification: Diagnostics

One optional summary is the `report` package's comparison output, including its approximate p-value; this should be treated as an aid to interpretation rather than a definitive model-selection test.

```
report::report(a)
```

The difference in predictive accuracy, as indexed by Expected Log Predictive Density (ELPD-LOO), suggests that 'model_dist_apim_long_complex' is the best model (ELPD = -14608.23), followed by 'model_dist_apim_long_simple' (diff-ELPD = -15.44 +- 6.68, p = 0.021).

In this instance, the maximal model (no AR) seems to be the preferred one.

In the case of overfitting, we could try to improve the model by subsequently excluding the smallest random effects (or correlations between them) and compare various models. For more guidance on random effects check out del Rosario & West (2025).

4.18 Maximal Model Specification: Comparing Estimates to Simulated Values (Ground Truth)

Group	Parameter	True (Simulated)	Estimate	95% CI	Coverage	Abs. Error	Recovery
Fixed	Intercept (male)	4.5659893	4.4841192	[4.313, 4.650]	Yes	0.082	Covered
Fixed	Intercept (female)	5.7823087	5.7407056	[5.544, 5.925]	Yes	0.042	Covered
Fixed	Time slope (male)	-0.0050000	-0.0030199	[-0.007, 0.001]	Yes	0.002	Covered
Fixed	Time slope (female)	0.0100000	0.0105530	[0.007, 0.014]	Yes	0.001	Covered
Between-person	Actor BP (male)	1.1855503	1.1106798	[0.895, 1.322]	Yes	0.075	Covered
Between-person	Partner BP (male)	0.2707719	0.3107712	[0.116, 0.504]	Yes	0.040	Covered
Between-person	Actor BP (female)	1.4298780	1.3773249	[1.151, 1.601]	Yes	0.053	Covered
Between-person	Partner BP (female)	0.5291829	0.5860368	[0.382, 0.801]	Yes	0.057	Covered
Within-person	Actor WP (male)	0.2000000	0.1816675	[0.137, 0.228]	Yes	0.018	Covered
Within-person	Partner WP (male)	0.1000000	0.1268520	[0.078, 0.174]	Yes	0.027	Covered
Within-person	Actor WP (female)	0.4000000	0.4206774	[0.391, 0.451]	Yes	0.021	Covered
Within-person	Partner WP (female)	0.2000000	0.1711715	[0.143, 0.199]	No	0.029	Miss
Random effects (SD)	Couple intercept SD (male)	0.7919045	0.8441943	[0.734, 0.972]	Yes	0.052	Covered
Random effects (SD)	Couple intercept SD (female)	0.8979616	0.9161830	[0.799, 1.057]	Yes	0.018	Covered
Random effects (SD)	Time slope SD (male)	0.0149848	0.0167148	[0.014, 0.020]	Yes	0.002	Covered
Random effects (SD)	Time slope SD (female)	0.0150338	0.0164663	[0.014, 0.019]	Yes	0.001	Covered
Random effects (SD)	Actor WP slope SD (male)	0.1006336	0.0893868	[0.017, 0.155]	Yes	0.011	Covered
Random effects (SD)	Partner WP slope SD (male)	0.0806085	0.1309919	[0.032, 0.202]	Yes	0.050	Covered
Random effects (SD)	Actor WP slope SD (female)	0.0955935	0.0776849	[0.022, 0.123]	Yes	0.018	Covered
Random effects (SD)	Partner WP slope SD (female)	0.0678586	0.0534853	[0.003, 0.104]	Yes	0.014	Covered
Random effects (cor)	cor(Intercept (male), Intercept (female))	0.1803231	0.1598557	[-0.039, 0.347]	Yes	0.020	Covered
Random effects (cor)	cor(Intercept (male), Time slope (male))	0.5083773	0.4711823	[0.281, 0.640]	Yes	0.037	Covered
Random effects (cor)	cor(Intercept (male), Time slope (female))	0.1921701	0.1744795	[-0.029, 0.365]	Yes	0.018	Covered
Random effects (cor)	cor(Intercept (male), Actor WP slope (male))	0.2262539	0.3521455	[-0.057, 0.702]	Yes	0.126	Covered
Random effects (cor)	cor(Intercept (male), Partner WP slope (male))	0.0953425	0.0632898	[-0.272, 0.400]	Yes	0.032	Covered
Random effects (cor)	cor(Intercept (male), Actor WP slope (female))	0.0398182	-0.0644726	[-0.421, 0.290]	Yes	0.104	Covered
Random effects (cor)	cor(Intercept (male), Partner WP slope (female))	0.3479821	0.2509883	[-0.224, 0.645]	Yes	0.097	Covered
Random effects (cor)	cor(Intercept (female), Time slope (male))	0.0526005	0.0108891	[-0.199, 0.236]	Yes	0.042	Covered
Random effects (cor)	cor(Intercept (female), Time slope (female))	0.6138595	0.4846139	[0.318, 0.635]	Yes	0.129	Covered
Random effects (cor)	cor(Intercept (female), Actor WP slope (male))	0.1319990	0.1910060	[-0.218, 0.572]	Yes	0.059	Covered
Random effects (cor)	cor(Intercept (female), Partner WP slope (male))	-0.0241164	0.1032397	[-0.241, 0.436]	Yes	0.127	Covered
Random effects (cor)	cor(Intercept (female), Actor WP slope (female))	0.1385319	0.1972091	[-0.171, 0.530]	Yes	0.059	Covered
Random effects (cor)	cor(Intercept (female), Partner WP slope (female))	-0.0263927	0.0596763	[-0.386, 0.505]	Yes	0.086	Covered
Random effects (cor)	cor(Time slope (male), Time slope (female))	0.1339757	-0.0301353	[-0.257, 0.197]	Yes	0.164	Covered
Random effects (cor)	cor(Time slope (male), Actor WP slope (male))	0.2400640	0.3041956	[-0.157, 0.679]	Yes	0.064	Covered
Random effects (cor)	cor(Time slope (male), Partner WP slope (male))	0.0918211	0.0471610	[-0.332, 0.418]	Yes	0.045	Covered
Random effects (cor)	cor(Time slope (male), Actor WP slope (female))	0.0024579	-0.1357687	[-0.512, 0.254]	Yes	0.138	Covered
Random effects (cor)	cor(Time slope (male), Partner WP slope (female))	0.2635152	0.1043193	[-0.356, 0.540]	Yes	0.159	Covered
Random effects (cor)	cor(Time slope (female), Actor WP slope (male))	0.0711586	0.2636537	[-0.168, 0.642]	Yes	0.192	Covered
Random effects (cor)	cor(Time slope (female), Partner WP slope (male))	0.0441926	0.1100475	[-0.243, 0.445]	Yes	0.066	Covered

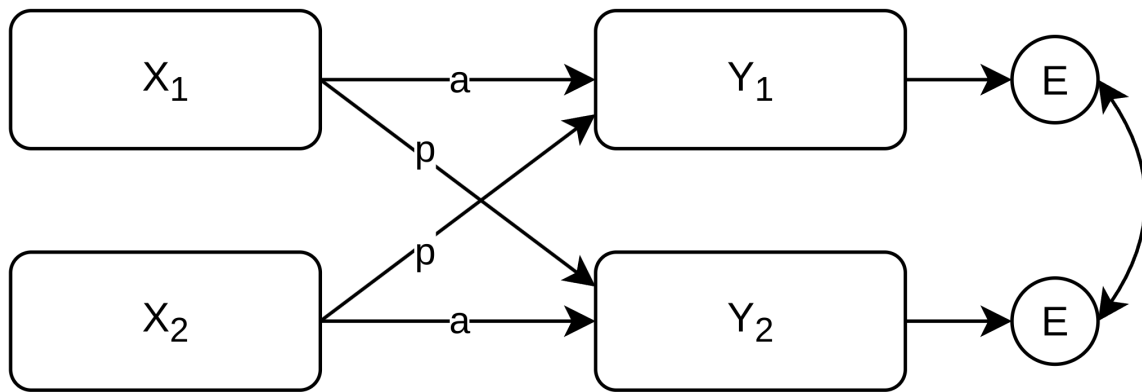
(continued)

Group	Parameter	True (Simulated)	Estimate	95% CI	Coverage	Abs. Error	Recovery
Random effects (cor)	cor(Time slope (female), Actor WP slope (female))	0.1380171	-0.0733653	[-0.421, 0.282]	Yes	0.211	Covered
Random effects (cor)	cor(Time slope (female), Partner WP slope (female))	0.0279326	0.1420481	[-0.353, 0.569]	Yes	0.114	Covered
Random effects (cor)	cor(Actor WP slope (male), Partner WP slope (male))	0.1128802	0.2377581	[-0.288, 0.679]	Yes	0.125	Covered
Random effects (cor)	cor(Actor WP slope (male), Actor WP slope (female))	0.0478755	-0.1413309	[-0.620, 0.406]	Yes	0.189	Covered
Random effects (cor)	cor(Actor WP slope (male), Partner WP slope (female))	0.1135981	0.1603468	[-0.409, 0.655]	Yes	0.047	Covered
Random effects (cor)	cor(Partner WP slope (male), Actor WP slope (female))	-0.0107839	-0.1047568	[-0.573, 0.383]	Yes	0.094	Covered
Random effects (cor)	cor(Partner WP slope (male), Partner WP slope (female))	0.1384787	0.0568673	[-0.471, 0.560]	Yes	0.082	Covered
Random effects (cor)	cor(Actor WP slope (female), Partner WP slope (female))	0.1494376	0.0851646	[-0.453, 0.600]	Yes	0.064	Covered
Residuals	Same-day residual correlation	0.0366176	0.0413669	[0.015, 0.068]	Yes	0.005	Covered
Residuals	Residual SD (male) (exp)	1.1210517	1.1150775	[1.094, 1.137]	Yes	0.006	Covered
Residuals	Residual SD (female) (exp)	0.7173353	0.7182000	[0.705, 0.732]	Yes	0.001	Covered

5 Exchangeable Dyads

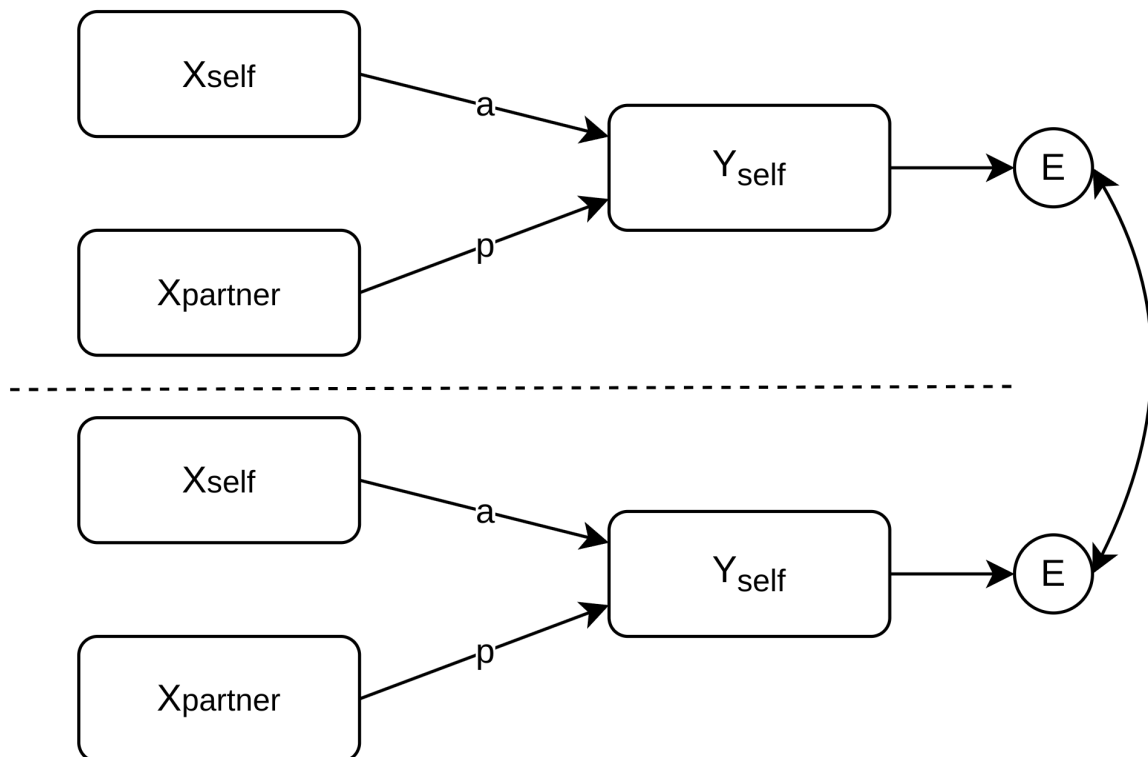
6 Exchangeable Dyads - Cross-Sectional

6.1 Exchangeable Dyads - Cross-Sectional APIM



(e.g., del Rosario & West, 2025; Kenny & Ackerman, 2023)

6.2 Exchangeable Dyads - Cross-Sectional APIM



(e.g., del Rosario & West, 2025; Kenny & Ackerman, 2023)

6.3 Exchangeable Dyads - Main Assumptions

Partners are exchangeable, i.e., not *systematically* different.

- Equal actor effects
- Equal partner effects
- Equal means
- Equal residual variances

But they should still be allowed to vary within each couple, while being correlated.

(e.g., del Rosario & West, 2025; Kenny & Ackerman, 2023)

6.4 Exchangeable Dyads - Cross-Sectional APIM: Data

Same cross-sectional data in the same 4-field actor-partner format as before. For illustration, we now fit exchangeability-constrained models to the same distinguishable heterosexual dyads; this is a modelling constraint/test, not a claim that the data-generating dyads are truly exchangeable.

```
print_apim_preview(df_apim)
```

userID	coupleID	member	gender	is male	is female	satisfaction	comm actor	comm partner
1 1	1	1	1	0	1	4.841332	-0.1553604	0.9980657
1 2	1	2	2	1	0	5.577165	0.9980657	-0.1553604
2 1	2	1	1	0	1	7.248741	0.8748537	1.4272563
2 2	2	2	2	1	0	6.960293	1.4272563	0.8748537
3 1	3	1	1	0	1	6.928699	-0.7224960	0.5095933
3 2	3	2	2	1	0	6.077688	0.5095933	-0.7224960

6.5 Exchangeable Dyads - Cross-Sectional APIM: glmmTMB

- For the fixed effects, we can pool the intercepts `is_male` and `is_female` by simply reintroducing a single shared intercept 1.
- For the actor and partner effects, we can remove the interactions to get a pooled estimate.
- For the residual, we want to constrain both partners' variances to be equal. Often, this is attempted via: `(1 | coupleID)`.
 - This gives both partners the same dyad-level random intercept variance.
 - With the usual Gaussian residual variance still present, the implied partner correlation is positive but not freely estimated: $\frac{\tau^2}{\tau^2 + \sigma^2}$.
 - If residual variance is removed with `dispformula = ~ 0`, the shared random intercept would imply a correlation near 1, which is too restrictive.

6.6 Exchangeable Dyads - Cross-Sectional APIM: glmmTMB

One solution in glmmTMB is the following:

```

1 model_exch_apim_glmmTMB <- glmmTMB(
2   satisfaction ~
3
4   # Pooled single intercept
5   1 +
6
7   # Pooled actor and partner effects
8   communication_actor_gmc + communication_partner_gmc +
9
10  # Homogeneous compound symmetry constrains the variances to be equal.
11  # we do not need to take gender here, we could construct an arbitrary categorization.
12  homcs(0 + is_male + is_female | coupleID)
13
14  # Since we model residuals via a cs random effects structure, glmmTMB requires us to remove an
15  # additional independent residual
16  , dispformula = ~ 0
17
18  , data = df_apim
19 )

```

6.7 Exchangeable Dyads - Cross-Sectional APIM: glmmTMB

This successfully returns a single pooled residual variance, and a correlation between partners:

```

Family: gaussian ( identity )
Formula:      satisfaction ~ 1 + communication_actor_gmc + communication_partner_gmc +
      homcs(0 + is_male + is_female | coupleID)
Dispersion:      ~0
Data: df_apim

      AIC      BIC    logLik -2*log(L)  df.resid
4026.6   4051.7   -2008.3   4016.6     1095

Random effects:

Conditional model:
Groups   Name    Variance Std.Dev. Corr
coupleID is_male 2.403    1.55    0.34 (cs)
Number of obs: 1100, groups: coupleID, 550

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    5.10934    0.05418   94.30 < 2e-16 ***
communication_actor_gmc 1.71990    0.03177   54.13 < 2e-16 ***
communication_partner_gmc 0.24137    0.03177    7.60 3.04e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

6.8 Exchangeable Dyads - Cross-Sectional APIM: glmmTMB

- The gender-coded cross-sectional version does not transfer directly to ILD.
- For repeated measures, we instead index the homogeneous two-partner residual structure at the couple-day level, as shown later.

6.9 Exchangeable Dyads - Cross-Sectional APIM: brms

In brms, this issue can be solved more easily with a native residual correlation structure and one pooled residual variance:

```

1 formula_exch_apim <- bf(
2   satisfaction ~ 1 +
3     communication_actor_gmc + communication_partner_gmc +
4
5   # To achieve a homogeneous two-partner covariance structure we use an
6   # unstructured residual correlation over the two member positions.
7   unstr(time = member, gr = coupleID)
8
9   # ... but now, we do not allow individual variances anymore.
10  # We want one pooled residual variance. So we do not specify a sigma formula.
11  # This implies:
12  # , sigma = ~ 1
13 )
14
15 priors_exch_apim <- c(
16   prior(normal(2, 10), class = "Intercept"),
17   prior(normal(0, 5), class = "b"),
18   prior(student_t(3, 0, 1.5), class = "sigma"),
19   prior(lkj(2), class = "cortime")
20 )
21
22 model_exch_apim <- brm(
23   formula = formula_exch_apim,
24   data = df_apim,
25   family = gaussian(link = identity),
26   prior = priors_exch_apim,
27   chains = 4,
28   cores = 4,
29   iter = 2000,
30   warmup = 1000,
31   seed = 123,
32   file = file.path('brms_cache', 'example1_ind_apim') # Cache the model
33 )

```

6.10 Exchangeable Dyads - Cross-Sectional APIM: Results

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
Fixed Effects					
Intercept	5.11	[5.00, 5.21]	1.001	4663	3034
comm actor	1.72	[1.66, 1.78]	1.001	4493	2982
comm partner	0.24	[0.18, 0.30]	1.001	5398	3170
Residual Structure					
cortime(1, 2)	0.34	[0.27, 0.41]	1.000	4188	2958
sigma	1.55	[1.49, 1.62]	1.000	4141	3044

6.11 Exchangeable Dyads - Cross-Sectional APIM: Test for Distinguishability

Leave-one-out (loo) cross-validation compares out-of-sample predictive accuracy. It can be meaningful for non-nested models when they are fit to the same outcome and observations, and when the pointwise log-likelihoods are comparable.

```

a <- loo_compare(
  loo(model_exch_apim),
  loo(model_dist_apim_b)
)
print(a)

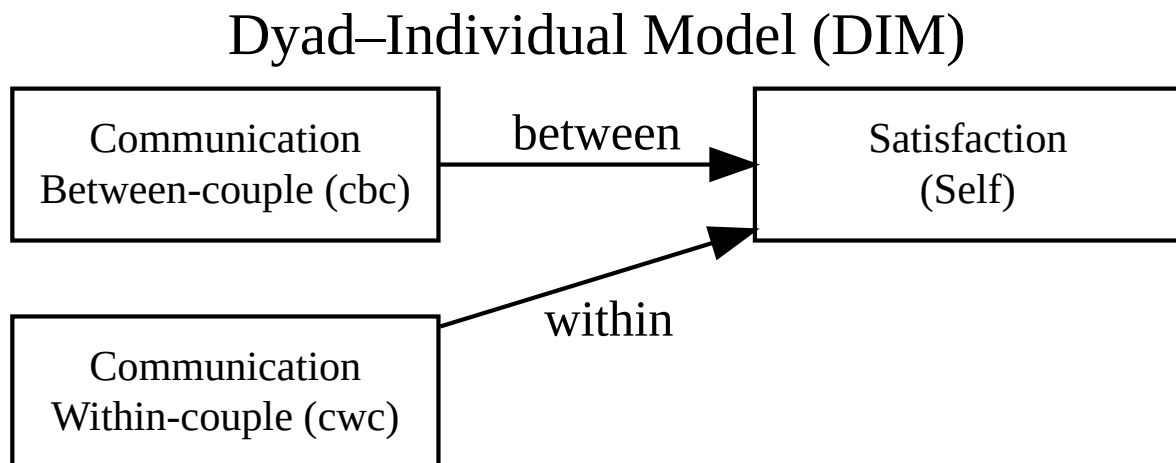
```

Model	elpd_diff	se_diff
model_dist_apim	0.0	0.0
model_exch_apim	-172.9	16.5

```
report::report(a)
```

The difference in predictive accuracy, as indexed by Expected Log Predictive Density (ELPD-LOO), suggests that ‘model_dist_apim_b’ is the best model (ELPD = -1806.42), followed by ‘model_exch_apim’ (diff-ELPD = -172.85 +- 16.53, $p < .001$). See the report package documentation for `report.compare.loo`.

6.12 Exchangeable Dyads - Cross-Sectional DIM



6.13 Exchangeable Dyads - Cross-Sectional DIM: Data

Starting from scratch (no 4-field data needed)

userID	coupleID	member	comm	satisfaction
1 1	1	1	4.851107	4.841332
1 2	1	2	6.004533	5.577165
2 1	2	1	5.881321	7.248741
2 2	2	2	6.433723	6.960293
3 1	3	1	4.283971	6.928699
3 2	3	2	5.516060	6.077688

6.14 Exchangeable Dyads - Cross-Sectional DIM: Centering

Decompose `communication` variance into:

- **Between-couple** (cbc): Couple-mean communication skills in relation to other couples
- **Within-couple** (cwc): Individuals’ communication skills in relation to their couple-mean

Same assumption about exchangeability as in the APIM

6.15 Exchangeable Dyads - Cross-Sectional DIM: Centering

```
df_dim <- df_apim2 %>%
  group_by(coupleID) %>%
  mutate(
    communication_cm = mean(communication, na.rm = TRUE),
    communication_cwc = communication - communication_cm
  ) %>%
  ungroup() %>%
  mutate(
    communication_cbc = communication_cm - mean(communication_cm, na.rm = TRUE)
  ) %>%
  dplyr::select(c('userID', 'coupleID', 'member', 'satisfaction', 'communication_cwc',
    ↪ 'communication_cbc'))
```

6.16 Exchangeable Dyads - Cross-Sectional DIM: Centering

userID	coupleID	member	satisfaction	comm Wc	comm Bc
1 1	1	1	4.841332	-0.5767130	0.4213527
1 2	1	2	5.577165	0.5767130	0.4213527
2 1	2	1	7.248741	-0.2762013	1.1510550
2 2	2	2	6.960293	0.2762013	1.1510550
3 1	3	1	6.928699	-0.6160447	-0.1064514
3 2	3	2	6.077688	0.6160447	-0.1064514

6.17 Exchangeable Dyads - Cross-Sectional DIM: Model

```
1 formula_exch_dim <- bf(
2   satisfaction ~ 1 +
3     communication_cbc + communication_cwc +
4     unstr(time = member, gr = coupleID)
5
6   # sigma ~ 1 is implied.
7 )
8
9 priors_exch_dim <- c(
10  prior(normal(2, 10), class = "Intercept"),
11  prior(normal(0, 5), class = "b"),
12  prior(student_t(3, 0, 1.5), class = "sigma"),
13  prior(lkj(2), class = "cortime")
14 )
15
16 model_exch_dim <- brm(
17   formula = formula_exch_dim,
18   data = df_dim,
19   family = gaussian(link = identity),
20   prior = priors_exch_dim,
21   chains = 4,
22   cores = 4,
23   iter = 2000,
24   warmup = 1000,
25   seed = 123,
26   file = file.path('brms_cache', 'example1_ind_dim') # Cache the model
27 )
```

6.18 Exchangeable Dyads - Cross-Sectional DIM: Results

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
Fixed Effects					

Intercept	5.11	[5.00, 5.22]	1.000	4517	3108
comm Bc	1.96	[1.88, 2.05]	1.005	4952	3056
comm Wc	1.48	[1.39, 1.57]	1.001	4418	3212
Residual Structure					
cortime(1, 2)	0.34	[0.27, 0.41]	1.002	4101	3331
sigma	1.55	[1.49, 1.62]	1.002	3981	3422

6.19 Equivalence APIM and DIM

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[5.00, 5.21]
comm actor	1.72	[1.66, 1.78]
comm partner	0.24	[0.18, 0.30]
Residual Structure		
cortime(1, 2)	0.34	[0.27, 0.41]
sigma	1.55	[1.49, 1.62]

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[5.00, 5.22]
comm Bc	1.96	[1.88, 2.05]
comm Wc	1.48	[1.39, 1.57]
Residual Structure		
cortime(1, 2)	0.34	[0.27, 0.41]
sigma	1.55	[1.49, 1.62]

$$b_{actor_gmc} + b_{partner_gmc} = b_{cbc}$$

$$b_{actor_gmc} - b_{partner_gmc} = b_{cwc}$$

(Bolger et al., 2025)

6.20 Equivalence APIM and DIM

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[5.00, 5.21]
comm actor	1.72	[1.66, 1.78]
comm partner	0.24	[0.18, 0.30]

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[5.00, 5.22]
comm Bc	1.96	[1.88, 2.05]
comm Wc	1.48	[1.39, 1.57]

In the HTML slides, this section contains an interactive slider demonstration. In the PDF, the same APIM-DIM reparameterization is shown statically:

- Actor and partner predictors can be rewritten as a dyad-level mean and a within-dyad difference.
- The DIM coefficients are a linear reparameterization of the APIM coefficients, so fitted values are unchanged when the same information is represented in either form.

```
x_cbc = (x_actor_gmc + x_partner_gmc) / 2
x_cwc = (x_actor_gmc - x_partner_gmc) / 2
```

6.21 Equivalence APIM and DIM

- If the couple mean goes up by 1 and the within-couple deviation from the couple mean stays fixed, this means that both partners' predictors must go up by 1 unit. Thus, the effects are linear combinations:

$$b_{cbc} = b_{actor_gmc} + b_{partner_gmc}$$

- Similarly, if a person's deviation from their couple mean is one unit higher and the couple mean stays fixed, this means their partners' deviation must be one unit lower. Thus, the effects are a linear combination:

$$b_{cwc} = b_{actor_gmc} - b_{partner_gmc}$$

(Bolger et al., 2025)

6.22 Equivalence APIM and DIM

- Conversely, to obtain the actor and partner effects given the DIM estimates:

$$b_{actor_gmc} = \frac{b_{cbc} + b_{cwc}}{2}$$

$$b_{partner_gmc} = \frac{b_{cbc} - b_{cwc}}{2}$$

6.23 Equivalence APIM and DIM

Example: using APIM coefficients to obtain DIM coefficients and computing *Credible Intervals* of DIM coefficients.

```
a <- hypothesis(
  model_exch_apim,
  "communication_actor_gmc + communication_partner_gmc = 0"
)

round(a$hypothesis[,c(2,3,4,5)], 2)
```

	Estimate	Est.Error	CI.Lower	CI.Upper
1	1.96	0.04	1.88	2.05

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[5.00, 5.21]
comm actor	1.72	[1.66, 1.78]
comm partner	0.24	[0.18, 0.30]

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[5.00, 5.22]
comm Bc	1.96	[1.88, 2.05]
comm Wc	1.48	[1.39, 1.57]

6.24 Equivalence APIM and DIM: Takeaway

- **APIM** and **DIM** are reparametrizations of the same model
- **APIM**: intuitive actor/partner framing
- **DIM**: clean separation of between vs within components
- Estimating one model allows for directly obtaining estimates of the other
- Same random-effects structure at dyad level and same assumptions

6.24.1 Side Note:

In the **distinguishable** case, the **Dyadic Score Model (DSM)** (e.g., Stadler et al., 2023) is equivalent to the APIM! (Bolger et al., 2025; Iida et al., 2018)

6.25 Exchangeable Dyads - L-DIM/L-APIM: Simulated Data

We use the same data as in the exchangeable case. But we don't need the gender-related variables anymore, as we assume that partners are indistinguishable.

userID	coupleID	couple day	day	day c	member	closeness	support actor	support partner	support actor BP	support actor WP
31 1	31	31.0	0	-27	1	5.47	6.02	7.09	0.65	0.41
31 1	31	31.1	1	-26	1	7.69	6.13	6.95	0.65	0.51
31 1	31	31.2	2	-25	1	5.83	6.1	6.34	0.65	0.49
31 1	31	31.3	3	-24	1	6.55	6.99	5.39	0.65	1.38
...	...	...	...	...	...	...	...	...	...	...
31 2	31	31.0	0	-27	2	5.19	7.09	6.02	0.58	1.54
31 2	31	31.1	1	-26	2	6.96	6.95	6.13	0.58	1.4
31 2	31	31.2	2	-25	2	6.9	6.34	6.1	0.58	0.79
31 2	31	31.3	3	-24	2	7.59	5.39	6.99	0.58	-0.15

7 Exchangeable Dyads - Intensive Longitudinal

7.1 Exchangeable Dyads - Longitudinal DIM/APIM

- We need an APIM or DIM on both the between person level and the within-person level.
- Due to equivalence, we could have a DIM on one level and an APIM on the other.

If we want a within-person DIM:

```
df_long_dim <- df_long_apim2 %>%
  mutate(

    # Between Person Level DIM (centering between couples and within couple between person)
    provided_support_cbc = (provided_support_actor_cbp + provided_support_partner_cbp) / 2,
    provided_support_cwcbp = (provided_support_actor_cbp - provided_support_partner_cbp) / 2,

    # Within Person Level DIM (couple mean and deviation on each day)
    provided_support_cwp_mean = (provided_support_actor_cwp + provided_support_partner_cwp) / 2,
    provided_support_cwp_halfdiff = (provided_support_actor_cwp - provided_support_partner_cwp) / 2
  )
```

7.2 Exchangeable Dyads - L-DIM/L-APIM: Preparing the sum and difference approach

We need to randomly assign -1 and +1 for each Partner within each couple. This will be needed for contrast coding the intercept.

For example:

```
df_long_dim <- df_long_dim %>%
  group_by(coupleID) %>%
  mutate(
    base = ifelse(userID == min(userID), 1, -1),
    flip = 1 - 2 * rbinom(1, 1, 0.5), # yields +1 or -1 once per couple
    Idiff = base * flip,

    # Not needed for brms, but for glmmTMB we create two dummy variables representing
    # randomly selected partner 1 and partner 2.
    is_partner1 = as.numeric(Idiff == 1),
    is_partner2 = as.numeric(Idiff == -1)
  ) %>%
  ungroup() %>%
  relocate(Idiff, .after = coupleID) %>%
  relocate(is_partner1, is_partner2, .after = Idiff) %>%
  dplyr::select(-base, -flip)
```

userID	coupleID	day	day c	member	support Bc	support Wcbp	support WP mean	support WP halfdiff
31 1	31	0	-27	1	0.61	0.04	0.97	-0.57
31 1	31	1	-26	1	0.61	0.04	0.96	-0.44
31 1	31	2	-25	1	0.61	0.04	0.64	-0.15
31 1	31	3	-24	1	0.61	0.04	0.61	0.77
...	...	...	...	...	...	...	...	...
31 2	31	0	-27	2	0.61	-0.04	0.97	0.57
31 2	31	1	-26	2	0.61	-0.04	0.96	0.44
31 2	31	2	-25	2	0.61	-0.04	0.64	0.15
31 2	31	3	-24	2	0.61	-0.04	0.61	-0.77

7.3 Exchangeable Dyads - L-DIM/L-APIM: Preparing the Sum and Difference Approach

userID	coupleID	Idiff	is partner1	is partner2	day	day c	support Bc	support Wcbp	support WP mean	support WP halfdiff
31 1	31	-1	0	1	0	-27	0.61	0.04	0.97	-0.57
31 1	31	-1	0	1	1	-26	0.61	0.04	0.96	-0.44
31 1	31	-1	0	1	2	-25	0.61	0.04	0.64	-0.15
31 1	31	-1	0	1	3	-24	0.61	0.04	0.61	0.77
31 1	31	-1	0	1	4	-23	0.61	0.04	0.36	0.72
...	...	...	...	...	...	...	...	...	...	...
31 2	31	1	1	0	0	-27	0.61	-0.04	0.97	0.57
31 2	31	1	1	0	1	-26	0.61	-0.04	0.96	0.44
31 2	31	1	1	0	2	-25	0.61	-0.04	0.64	0.15
31 2	31	1	1	0	3	-24	0.61	-0.04	0.61	-0.77

7.4 Exchangeable Dyads - L-DIM/L-APIM: glmmTMB

```

1 model_exch_long_apim_glmmTMB <- glmmTMB(
2   closeness ~ 1 +
3
4   diaryday_c +
5
6   # Within-person APIM
7   provided_support_actor_cwp +
8   provided_support_partner_cwp +
9   # Equivalent to within-person DIM:
10  # provided_support_cwp_mean +
11  # provided_support_cwp_halfdiff +
12
13  # Between-person APIM
14  provided_support_actor_cbp +
15  provided_support_partner_cbp +
16  # Equivalent to between-person DIM
17  # provided_support_cwcbp+
18  # provided_support_cbc +
19
20  # Dyad-Level intercept and slopes for time-varying predictors
21  (1 + diaryday_c + provided_support_actor_cwp + provided_support_partner_cwp | coupleID) +
22
23  # Both partners' deviations from these dyad-level means and slopes.
24  # Note: separate random effects block to make them uncorrelated form dyad-level REs.
25  (0 + Idiff +
26    I(Idiff * diaryday_c) +
27    I(Idiff * provided_support_actor_cwp) +
28    I(Idiff * provided_support_partner_cwp) | coupleID) +
29
30  # After accounting for stable differences between partners, they may still be
31  # non-independent each day due to external occurrences and daily exchange processes
32  # not captured in the fixed effects. Thus, residuals are not yet independent.
33  # Homogeneous same-day residual covariance between partners:
34  homcs(0 + is_partner1 + is_partner2 | coupleID:diaryday)
35
36  # We remove the additional independent residual variance
37  # (only if the family has a native residual!)
38  , dispformula = ~ 0
39  , data = df_long_dim
40 )

```

7.5 Exchangeable Dyads - L-DIM/L-APIM: glmmTMB

```

Family: gaussian ( identity )
Formula:      closeness ~ 1 + diaryday_c + provided_support_actor_cwp + provided_support_partner_cwp +
↪ provided_support_actor_cbp + provided_support_partner_cbp +
(1 + diaryday_c + provided_support_actor_cwp + provided_support_partner_cwp |
coupleID) + (0 + Idiff + I(Idiff * diaryday_c) + I(Idiff *
provided_support_actor_cwp) + I(Idiff * provided_support_partner_cwp) |
coupleID) + homcs(0 + is_partner1 + is_partner2 | coupleID:diaryday)
Dispersion:      ~0
Data: df_long_dim

```

AIC	BIC	logLik	-2*log(L)	df.resid
NA	NA	NA	NA	10972

Random effects:

Conditional model:

Groups	Name	Variance	Std.Dev.	Corr
coupleID	(Intercept)	0.4295833	0.65543	

```

      diaryday_c                0.0001292 0.01137 0.58
      provided_support_actor_cwp 0.0011353 0.03369 0.67      0.99
      provided_support_partner_cwp 0.0077910 0.08827 0.20      0.17 0.19
coupleID.1 Idiff                0.7223445 0.84991
      I(Idiff * diaryday_c)      0.0001757 0.01325 0.64
      I(Idiff * provided_support_actor_cwp) 0.0194663 0.13952 0.73      0.48
      I(Idiff * provided_support_partner_cwp) 0.0050428 0.07101 0.15      0.14 0.45
coupleID.diaryday is_partner1 0.8770507 0.93651 0.04 (cs)
Number of obs: 11000, groups: coupleID, 100; coupleID:diaryday, 5500

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    5.109675   0.066172   77.22 < 2e-16 ***
diaryday_c      0.003769   0.001276    2.95 0.00313 **
provided_support_actor_cwp 0.299840   0.012682   23.64 < 2e-16 ***
provided_support_partner_cwp 0.149087   0.015068    9.89 < 2e-16 ***
provided_support_actor_cbp  1.307265   0.083097   15.73 < 2e-16 ***
provided_support_partner_cbp 0.397499   0.083031    4.79 1.69e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

In this example, the glmmTMB model does not converge properly. The random-effects structure is too complex for the information available in these data, so we would have to simplify it. Instead, we use brms below.

7.6 Exchangeable Dyads - L-DIM/L-APIM: brms

In brms, the same model can be estimated as:

Unlike the distinguishable longitudinal model, this exchangeable specification keeps one pooled residual `sigma` (`sigma ~ 1` is implied). Its residual SD prior therefore targets `class = "sigma"` directly.

```

1 formula_exch_long_apim <- bf(
2   closeness ~ 1 +
3
4   diaryday_c +
5
6   # Within-person APIM
7   provided_support_actor_cwp +
8   provided_support_partner_cwp +
9
10  # Between-person APIM
11  provided_support_actor_cbp +
12  provided_support_partner_cbp +
13
14  # Dyad-Level intercept and slopes for time-varying predictors
15  (1 + diaryday_c + provided_support_actor_cwp + provided_support_partner_cwp | coupleID) +
16
17  # Both partners' deviations from these dyad-level means and slopes.
18  # Note: separate random effects block to make them uncorrelated from dyad-level REs.
19  (0 + Idiff +
20    I(Idiff * diaryday_c) +
21    I(Idiff * provided_support_actor_cwp) +
22    I(Idiff * provided_support_partner_cwp) | coupleID) +
23
24  # Modelling Level 1 residuals in brms:
25  unstr(time = member, gr = couple_day)
26
27  # Again, no need to model heterogeneous residual variances (sigma)
28  # Implied: sigma = ~ 1
29 )
30
31 priors_exch_long_apim <- c(
32   prior(normal(4, 2), class = "Intercept"),
33   prior(normal(0, 2), class = "b"),
34   prior(exponential(1), class = "sd"),

```

```

35 prior(lkj(2), class = "cor"),
36 prior(student_t(3, 0, 1.5), class = "sigma"),
37 prior(lkj(2), class = "cortime")
38 )
39
40 model_exch_long_apim <- brm(
41   formula = formula_exch_long_apim,
42   data = df_long_dim,
43   family = gaussian(link = identity),
44   prior = priors_exch_long_apim,
45   chains = 4,
46   cores = 4,
47   iter = 2000,
48   warmup = 1000,
49   seed = 123,
50   file = file.path('brms_cache', 'model_apim_ind_long_apim')
51 )

```

7.7 Exchangeable Dyads - L-DIM/L-APIM: brms

The equivalent DIM specification:

```

1 # DIM Version
2 formula_exch_long_dim <- bf(
3   closeness ~ 1 +
4
5     diaryday_c +
6
7     provided_support_cwp_mean +
8     provided_support_cwp_halfdiff +
9
10    provided_support_cbc +
11    provided_support_cwcbp +
12
13    (1 + diaryday_c + provided_support_actor_cwp + provided_support_partner_cwp | coupleID) +
14    (0 + Idiff +
15      I(Idiff * diaryday_c) +
16      I(Idiff * provided_support_actor_cwp) +
17      I(Idiff * provided_support_partner_cwp) | coupleID) +
18
19    unstr(time = member, gr = couple_day)
20 )
21
22 priors_exch_long_dim <- c(
23   prior(normal(4, 2), class = "Intercept"),
24   prior(normal(0, 2), class = "b"),
25   prior(exponential(1), class = "sd"),
26   prior(lkj(2), class = "cor"),
27   prior(student_t(3, 0, 1.5), class = "sigma"),
28   prior(lkj(2), class = "cortime")
29 )
30
31 model_exch_long_dim <- brm(
32   formula = formula_exch_long_dim,
33   data = df_long_dim,
34   family = gaussian(link = identity),
35   prior = priors_exch_long_dim,
36   chains = 4,
37   cores = 4,
38   iter = 2000,
39   warmup = 1000,
40   seed = 123,
41   file = file.path('brms_cache', 'model_apim_ind_long_dim')
42 )

```

7.8 Exchangeable Dyads - L-DIM/L-APIM: Explanation Idiff

For appropriate random effects, we can use the Sum and Difference approach (del Rosario & West, 2025; Kenny & Ackerman, 2023):

- Randomly give one partner a constant of -1 and the other a constant of 1
- The couple-level intercept represents the average of the two partners' intercepts; the corresponding sum would be twice this value. (1 | coupleID)
- A column of 1s and -1s represent deviations (difference) of each partner from the couple intercept with each partner contributing equally in one direction (+1 and -1)

Exchangeability is retained: if partners are flipped, the result is the same.

(del Rosario & West (2025) provide practical guidance on how to reduce the random effects structure in case of non-convergence.)

7.9 Exchangeable Dyads - L-DIM/L-APIM: homcs and Idiff

For two partners coded with Idiff = +1 and Idiff = -1, the glmmTMB residual structures `homcs(0 + is_partner1 + is_partner2 | coupleID:diaryday)` and `(1 | coupleID:diaryday) + (0 + Idiff | coupleID:diaryday)` imply the same homogeneous two-partner residual covariance.

This equivalence assumes exactly two rows per couple-day, independent sum and difference random-effect blocks, and `dispformula = ~ 0`.

The model code keeps `homcs(...)` because it states the residual structure more directly.

It does also not need back-transformation.

7.10 Exchangeable Dyads - L-DIM/L-APIM: Results

Comparing APIM (left) and DIM (right) output of fixed effects.

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[4.99, 5.23]
day c	0.00	[0.00, 0.01]
support actor WP	0.30	[0.27, 0.33]
support partner WP	0.15	[0.12, 0.18]
support actor BP	1.30	[1.14, 1.48]
support partner BP	0.40	[0.23, 0.56]

	Est.	95% CI
Fixed Effects		
Intercept	5.11	[4.99, 5.24]
day c	0.00	[0.00, 0.01]
support WP mean	0.45	[0.41, 0.49]
support WP halfdiff	0.15	[0.11, 0.19]
support Bc	1.71	[1.56, 1.87]
support Wcbp	0.91	[0.59, 1.21]

- Equivalence holds on both levels. Numeric deviations arise due to priors, sampling noise, and rounding.

7.11 Exchangeable Dyads - L-DIM/L-APIM: Results

APIM Results in Detail:

	Est.	95% CI	Rhat	Bulk ESS	Tail ESS
Fixed Effects					
Intercept	5.11	[4.99, 5.23]	1.018	215	329
day c	0.00	[0.00, 0.01]	1.003	767	1940
support actor WP	0.30	[0.27, 0.33]	1.002	2795	2545
support partner WP	0.15	[0.12, 0.18]	1.001	2728	2805
support actor BP	1.30	[1.14, 1.48]	1.012	253	543
support partner BP	0.40	[0.23, 0.56]	1.018	235	474
Random Effects					
sd(Intercept)	0.67	[0.58, 0.78]	1.009	345	714
sd(day c)	0.01	[0.01, 0.01]	1.002	1044	2098
sd(support actor WP)	0.05	[0.01, 0.09]	1.013	499	378
sd(support partner WP)	0.08	[0.04, 0.12]	1.002	1159	1274
sd(Idiff)	0.85	[0.74, 0.97]	1.039	150	350
sd(IIdiffMUday c)	0.01	[0.01, 0.02]	1.005	671	1470
sd(IIdiffMUsupport actor WP)	0.14	[0.11, 0.18]	1.003	1291	2287
sd(IIdiffMUsupport partner WP)	0.06	[0.01, 0.11]	1.002	625	649
cor(Intercept, day c)	0.56	[0.38, 0.70]	1.001	922	2124
cor(Intercept, support actor WP)	0.48	[-0.03, 0.83]	1.002	2251	1200
cor(day c, support actor WP)	0.37	[-0.15, 0.77]	1.001	2184	2399
cor(Intercept, support partner WP)	0.21	[-0.12, 0.52]	1.002	3523	2296
cor(day c, support partner WP)	0.19	[-0.17, 0.54]	1.000	2773	2715
cor(support actor WP, support partner WP)	0.31	[-0.30, 0.80]	1.004	564	1156
cor(Idiff, IIdiffMUday c)	0.60	[0.45, 0.72]	1.007	836	1789
cor(Idiff, IIdiffMUsupport actor WP)	0.66	[0.45, 0.81]	1.002	1713	2750
cor(IIdiffMUday c, IIdiffMUsupport actor WP)	0.41	[0.14, 0.63]	1.004	1321	2396
cor(Idiff, IIdiffMUsupport partner WP)	0.13	[-0.30, 0.55]	1.001	2681	1660
cor(IIdiffMUday c, IIdiffMUsupport partner WP)	0.12	[-0.35, 0.55]	1.001	2179	1510
cor(IIdiffMUsupport actor WP, IIdiffMUsupport partner WP)	0.36	[-0.16, 0.77]	1.001	1796	2341
cortime(1, 2)	0.04	[0.01, 0.06]	1.000	4708	2973
Residual Structure					
sigma	0.94	[0.92, 0.95]	1.000	4221	3007

7.12 Extract Full APIM Random Effects Variance-Covariance Matrix

We can rotate the random effects structure back to obtain the full APIM Random effects variance-covariance matrix, based on (Kenny & Ackerman, 2023). Just as we would in a SEM model with equality constraints!

For example, the within-person variance of the intercept is:

$$var(Intercept) + var(Idiff)$$

and the cross-person covariance of both partners' intercepts is:

$$var(Intercept) - var(Idiff)$$

Note: This only holds true when the means and Idiffs are uncorrelated. This is why we put them in separate random effects blocks.

7.13 Extract Full APIM Random Effects Variance-Covariance Matrix

```
# Custom function
rotate_apim_covariance <- function(
  fit,
  gr = "coupleID",
  Idiff = "Idiff"
) {
  random_effects <- fit$ranef
  random_effects <- random_effects[random_effects$group == gr,]

  SUM <- random_effects$coef[!grepl(Idiff, random_effects$coef)]
  DIFF <- random_effects$coef[grepl(Idiff, random_effects$coef)]

  varcor <- VarCorr(fit)[[gr]]

  within_person_covariance_matrix <- varcor$cov[SUM,'Estimate',SUM] + varcor$cov[DIFF,'Estimate',DIFF]
  cross_person_covariance_matrix <- varcor$cov[SUM,'Estimate',SUM] - varcor$cov[DIFF,'Estimate',DIFF]

  p <- nrow(within_person_covariance_matrix)

  Full <- rbind(
    cbind(within_person_covariance_matrix, cross_person_covariance_matrix),
    cbind(cross_person_covariance_matrix, within_person_covariance_matrix)
  )

  # nice labels
  base <- SUM
  rn <- c(paste0("PartnerA_", base), paste0("PartnerB_", base))
  colnames(Full) <- rn
  rownames(Full) <- rn

  return(list(within_person_covariance_matrix=within_person_covariance_matrix,
    ↪ cross_person_covariance_matrix=cross_person_covariance_matrix, full_covariance_matrix=Full))
}

a <- rotate_apim_covariance(model_exch_long_apim)
```

7.14 Extract Full APIM Random Effects Variance-Covariance Matrix: Within-Person Matrix

```
print_df(a$within_person_covariance_matrix)
```

	Intercept	day c	support actor WP	support partner WP
Intercept	1.1762392	0.0111609	0.0950373	0.0181542
day c	0.0111609	0.0003154	0.0009900	0.0002800
support actor WP	0.0950373	0.0009900	0.0237272	0.0042447
support partner WP	0.0181542	0.0002800	0.0042447	0.0119029

7.15 Extract Full APIM Random Effects Variance-Covariance Matrix: Cross-Person Matrix

```
print_df(a$cross_person_covariance_matrix)
```

	Intercept	day c	support actor WP	support partner WP
Intercept	-0.2746323	-0.0024732	-0.0635502	0.0044695
day c	-0.0024732	-0.0000422	-0.0005675	0.0000869
support actor WP	-0.0635502	-0.0005675	-0.0179524	-0.0017628
support partner WP	0.0044695	0.0000869	-0.0017628	0.0028713

7.16 Extract Full APIM Random Effects Variance-Covariance Matrix: Full Matrix

```
print_wide_matrix(a$full_covariance_matrix, font_size = 6)
```

	PA Int	PA day c	PA sup act cwp	PA sup prt cwp	PB Int	PB day c	PB sup act cwp	PB sup prt cwp
PA Int	1.176	0.011	0.095	0.018	-0.275	-0.002	-0.064	0.004
PA day c	0.011	0.000	0.001	0.000	-0.002	0.000	-0.001	0.000
PA sup act cwp	0.095	0.001	0.024	0.004	-0.064	-0.001	-0.018	-0.002
PA sup prt cwp	0.018	0.000	0.004	0.012	0.004	0.000	-0.002	0.003
PB Int	-0.275	-0.002	-0.064	0.004	1.176	0.011	0.095	0.018
PB day c	-0.002	0.000	-0.001	0.000	0.011	0.000	0.001	0.000
PB sup act cwp	-0.064	-0.001	-0.018	-0.002	0.095	0.001	0.024	0.004
PB sup prt cwp	0.004	0.000	-0.002	0.003	0.018	0.000	0.004	0.012

7.17 Convert to correlation matrix of random effects

```
# Custom function
sd_cor <- function(Sigma) {
  sds <- sqrt(diag(Sigma))
  cors <- cov2cor(Sigma)
  list(sd = sds, cor = cors)
}

full <- sd_cor(a$full_covariance_matrix)
```

7.18 Random effects SDs

```
print_df(as.data.frame(round(full$sd, 3)))
```

round(full\$sd, 3)	
PartnerA Intercept	1.085
PartnerA day c	0.018

PartnerA support actor WP	0.154
PartnerA support partner WP	0.109
PartnerB Intercept	1.085
PartnerB day c	0.018
PartnerB support actor WP	0.154
PartnerB support partner WP	0.109

7.19 Random effect correlation matrix

```
print_wide_matrix(full$cor, font_size = 6)
```

	PA Int	PA day c	PA sup act cwp	PA sup prt cwp	PB Int	PB day c	PB sup act cwp	PB sup prt cwp
PA Int	1.000	0.579	0.569	0.153	-0.233	-0.128	-0.380	0.038
PA day c	0.579	1.000	0.362	0.144	-0.128	-0.134	-0.207	0.045
PA sup act cwp	0.569	0.362	1.000	0.253	-0.380	-0.207	-0.757	-0.105
PA sup prt cwp	0.153	0.144	0.253	1.000	0.038	0.045	-0.105	0.241
PB Int	-0.233	-0.128	-0.380	0.038	1.000	0.579	0.569	0.153
PB day c	-0.128	-0.134	-0.207	0.045	0.579	1.000	0.362	0.144
PB sup act cwp	-0.380	-0.207	-0.757	-0.105	0.569	0.362	1.000	0.253
PB sup prt cwp	0.038	0.045	-0.105	0.241	0.153	0.144	0.253	1.000

References

- Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3), 255–278. <https://doi.org/10.1016/j.jml.2012.11.001>
- Bolger, N., Laurenceau, J.-P., & DiGiovanni, A. (2025). Unified analysis model for indistinguishable and distinguishable dyads. *Innovations in Interpersonal Relationships and Health Research: Advancing the Integration of Interdisciplinary Approaches to Dyadic Behavior Change*. <https://doi.org/10.17605/OSF.IO/WYDCJ>
- Bürkner, P.-C. (2025). *Brms: Bayesian regression models using stan*. <https://github.com/paul-buerkner/brms>
- del Rosario, K. S., & West, T. V. (2025). A Practical Guide to Specifying Random Effects in Longitudinal Dyadic Multilevel Modeling. *Advances in Methods and Practices in Psychological Science*, 8(3), 25152459251351286. <https://doi.org/10.1177/25152459251351286>
- Gabry, J., Češnovar, R., Johnson, A., & Bröder, S. (2025). *Cmdstanr: R interface to CmdStan*. <https://mc-stan.org/cmdstanr/>
- Guo, J., Gabry, J., Goodrich, B., Johnson, A., Weber, S., & Badr, H. S. (2025). *Rstan: R interface to stan*. <https://mc-stan.org/rstan/>
- Hartig, F. (2026). *DHARMA: Residual diagnostics for hierarchical (multi-level / mixed) regression models*. <http://florianhartig.github.io/DHARMA/>
- Iida, M., Seidman, G., & Shrout, P. E. (2018). Models of interdependent individuals versus dyadic processes in relationship research. *Journal of Social and Personal Relationships*, 35(1), 59–88. <https://doi.org/10.1177/0265407517725407>
- Kenny, D. A., & Ackerman, R. A. (2023). *Estimation of random effects with over-time dyadic data using multilevel modeling: The sum and difference method*. OSF Preprints. <https://doi.org/10.31219/osf.io/fju72>
- Kenny, D. A., & Cook, W. (1999). Partner effects in relationship research: Conceptual issues, analytic difficulties, and illustrations. *Personal Relationships*, 6(4), 433–448. <https://doi.org/10.1111/j.1475-6811.1999.tb00202.x>
- Kenny, D. A., & Kashy, D. A. (2011). Dyadic Data Analysis Using Multilevel Modeling. In *Handbook of Advanced Multilevel Analysis*. Routledge.
- Kenny, D. A., Kashy, D. A., & Cook, W. L. (2006). *Dyadic data analysis*. Guilford Press.
- Lüdtke, D., Makowski, D., Ben-Shachar, M. S., Patil, I., Wiernik, B. M., Bacher, E., & Thériault, R. (2026). *Easystats: Framework for easy statistical modeling, visualization, and reporting*. <https://easystats.github.io/easystats/>
- Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica*, 49(6), 1417–1426. <https://doi.org/10.2307/1911408>
- Pinheiro, J., Bates, D., & R Core Team. (2026). *Nlme: Linear and nonlinear mixed effects models*. <https://svn.r-project.org/R-packages/trunk/nlme/>
- Ripley, B., & Venables, B. (2025). *MASS: Support functions and datasets for venables and ripley's MASS*. <http://www.stats.ox.ac.uk/pub/MASS4/>
- Stadler, G., Scholz, U., Bolger, N., Shrout, P. E., Knoll, N., & Lüscher, J. (2023). How is companionship related to romantic partners' affect, relationship satisfaction, and health behavior? Using a longitudinal dyadic score model to understand daily and couple-level effects of a dyadic predictor. *Applied Psychology: Health and Well-Being*, 15(4), 1530–1554. <https://doi.org/10.1111/aphw.12450>
- Vehtari, A., Gabry, J., Magnusson, M., Yao, Y., Bürkner, P.-C., Paananen, T., & Gelman, A. (2025). *Loo: Efficient leave-one-out cross-validation and WAIC for bayesian models*. <https://mc-stan.org/loo/>
- Vuorre, M. (2023). *Bmlm: Bayesian multilevel mediation*. <https://github.com/mvuorre/bmlm/>
- Wickham, H. (2023). *Tidyverse: Easily install and load the tidyverse*. <https://tidyverse.tidyverse>

